Ambient air quality — Standard method for the measurement of the concentration of carbon monoxide by nondispersive infrared spectroscopy

The European Standard EN 14626:2005 has the status of a British Standard

ICS 13.040.20

National foreword

This British Standard is the official English language version of EN 14626:2005.

The UK participation in its preparation was entrusted by Technical Committee EH/2, Air quality, to Subcommittee EH/2/3, Ambient atmospheres, which has the responsibility to:

— aid enquirers to understand the text;

— present to the responsible international/European committee any enquiries on the interpretation, or proposals for change, and keep the UK interests informed;

— monitor related international and European developments and promulgate them in the UK.

A list of organizations represented on this subcommittee can be obtained on request to its secretary.

Cross-references

The British Standards which implement international or European publications referred to in this document may be found in the *BSI Catalogue* under the section entitled "International Standards Correspondence Index", or by using the "Search" facility of the *BSI Electronic Catalogue* or of British Standards Online.

This publication does not purport to include all the necessary provisions of a contract. Users are responsible for its correct application.

Compliance with a British Standard does not of itself confer immunity from legal obligations.

This British Standard, was published under the authority of the Standards Policy and Strategy Committee on 31 March 2005

Summary of pages

This document comprises a front cover, an inside front cover, the EN title page, pages 2 to 86, an inside back cover and a back cover.

The BSI copyright notice displayed in this document indicates when the document was last issued.

Amendments issued since publication

Amd. No.	Date	Comments

ISBN 0 580 45718 4

EUROPEAN STANDARD

NORME EUROPÉENNE

EUROPÄISCHE NORM

EN 14626

March 2005

ICS 13.040.20

English version

Ambient air quality - Standard method for the measurement of the concentration of carbon monoxide by nondispersive infrared spectroscopy

Qualité de l'air ambiant - Méthode normalisée de mesurage de la concentration en monoxyde de carbone par la méthode à rayonnement infrarouge non-dispersif

Luftqualität - Messverfahren zur Bestimmung von Kohlenmonoxid in Luft mit dem NDIR-Verfahren

This European Standard was approved by CEN on 10 December 2004.

CEN members are bound to comply with the CEN/CENELEC Internal Regulations which stipulate the conditions for giving this European Standard the status of a national standard without any alteration. Up-to-date lists and bibliographical references concerning such national standards may be obtained on application to the Central Secretariat or to any CEN member.

This European Standard exists in three official versions (English, French, German). A version in any other language made by translation under the responsibility of a CEN member into its own language and notified to the Central Secretariat has the same status as the official versions.

CEN members are the national standards bodies of Austria, Belgium, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Slovakia, Slovenia, Spain, Sweden, Switzerland and United Kingdom.

EUROPEAN COMMITTEE FOR STANDARDIZATION
COMITÉ EUROPÉEN DE NORMALISATION
EUROPÄISCHES KOMITEE FÜR NORMUNG

Management Centre: rue de Stassart, 36 B-1050 Brussels

Ref. No. EN 14626:2005: E

Contents

Page

Foreword 4

1 Scope 5

2 Normative references 5

3 Terms and definitions 6

4 Symbols and abbreviated terms 10

5 Principle 13
5.1 General 13
5.2 Measuring principle 13
5.3 Type approval test 13
5.4 Field operation and quality control 14

6 Sampling equipment 14
6.1 General 14
6.2 Sampling location 14
6.3 Sample inlet and sampling line 14
6.4 Particulate filter 15
6.5 Control and regulation of sample flow rate 15
6.6 Sampling pump for the manifold 15

7 Analyser equipment 16
7.1 General 16
7.2 Interferences due to infrared absorbing gases 16
7.3 Details about analyser equipment 16
7.4 Pressure measurement 17
7.5 Flow rate indicator 17
7.6 Sampling pump for the analyser 17

8 Type approval of CO-NDIR analysers 17
8.1 General 17
8.2 Relevant performance characteristics and performance criteria 17
8.3 Design changes 19
8.4 Procedures for determination of the performance characteristics during the laboratory test 19
8.5 Determination of the performance characteristics during the field test 29
8.6 Expanded uncertainty calculation for type approval 32

9 Field operation and ongoing quality control 33
9.1 General 33
9.2 Suitability evaluation 33
9.3 Initial installation 34
9.4 Ongoing quality control 35
9.5 Calibration of the analyser 36
9.6 Checks 37
9.7 Maintenance 40
9.8 Data handling and data reports 41

10 Expression of results 41

11 Test reports and documentation 42
11.1 Type approval tests 42
11.2 Field operation 42

Annex A (normative) Test of lack of fit 43

Annex B (informative) Sampling equipment 45

Annex C (informative) Sampling on micro scale 47
Annex D (informative) Schematics of non-dispersive infrared spectrometer 48
Annex E (informative) Manifold testing equipment 50
Annex F (normative) Type approval 51
Annex G (normative) Calculation of uncertainty in field operation at the 8-hour mean limit value 73
Bibliography 86

Foreword

This document (EN 14626:2005) has been prepared by Technical Committee CEN/TC 264 “Air quality”, the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by September 2005, and conflicting national standards shall be withdrawn at the latest by September 2005.

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Slovakia, Slovenia, Spain, Sweden, Switzerland and United Kingdom.

1 Scope

This document specifies a continuous measurement method for the determination of the concentration of carbon monoxide present in ambient air based on the non-dispersive infrared measuring principle (NDIR). This document describes the performance characteristics and sets the relevant minimum criteria required to select an appropriate non-dispersive infrared carbon monoxide analyser by means of type approval tests. It also includes the evaluation of the suitability of an anayser for use in a specific fixed site so as to meet the Directives data quality requirements and requirements during sampling, calibration and quality assurance.

The method is applicable to the determination of the mass concentration of carbon monoxide present in ambient air in the range from 0 mg/m^3 to 100 mg/m^3 carbon monoxide. This concentration range represents the certification range for the type approval test.

NOTE 1 0 mg/m^3 to 100 mg/m^3 of CO corresponds to 0 µmol/mol to 86 µmol/mol of CO.

The method covers the determination of ambient air concentrations of carbon monoxide in zones classified as rural areas, urban-background areas and traffic-orientated locations.

NOTE 2 Other ranges may be used for measurement systems applied at rural locations monitoring Ecosystems.

The results are expressed in mg/m^3 (at 293 K and 101,3 kPa).

When the standard is used for other purposes than the EU directive, the range and uncertainty requirements need not apply.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ENV 13005, *Guide to the expression of uncertainty in measurement*

EN ISO 14956:2002, *Air quality — Evaluation of the suitability of a measurement procedure by comparison with a required measurement uncertainty (ISO 14956:2002)*

ISO 6143, *Gas analysis — Comparison methods for determining and checking the composition of calibration gas mixtures*

ISO 6144, *Gas analysis — Preparation of calibration gas mixtures — Static volumetric method*

ISO 6145 (all parts), *Gas analysis — Preparation of calibration gas mixtures using dynamic volumetric methods*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

3.1
ambient air
outdoor air in the troposphere excluding workplace air[1)]

3.2
ambient temperature
temperature at the sample inlet outside the monitoring station (sample temperature, outdoor temperature)

3.3
availability of the analyser
fraction of the total time period for which usually valid measuring data of the ambient air concentration is available from an analyser

3.4
certification range
concentration range for which the analyser is type approved

3.5
calibration
comparison of the analyser response to a known gas concentration with a known uncertainty

3.6
combined standard uncertainty
calculation result of combining the uncertainties determined from all performance characteristics specified in this document according to the prescribed procedures given in this document

3.7
coverage factor
numerical factor used as a multiplier of the combined standard uncertainty in order to obtain an expanded uncertainty

3.8
designated body
body which has been designated for a specific task (type approval tests and/or QA/QC activities in the field) by the competent authority in the Member States

NOTE It is recommended that the designated body is accredited for the specific task according to EN ISO/IEC 17025.

3.9
expanded uncertainty
quantity defining an interval about the result of a measurement that may be expected to encompass a large fraction of the distribution of values that could reasonably be attributed to the measurand

NOTE for the purpose of this document the expanded uncertainty is the combined standard uncertainty multiplied by a coverage factor k = 2 resulting in an interval with a level of confidence of 95 %

3.10
fall time
difference between the response time (fall) and the lag time (fall)

[1)] As stated in the relevant EU legislation.

3.11
independent measurement
individual measurement that is not influenced by a previous individual measurement by separating two individual measurements by at least four response times

3.12
individual measurement
measurement averaged over a time period equal to the response time of the analyser

3.13
influence quantity
quantity that is not the measurand but that affects the result of the measurement (VIM 2.7), either an interferent influence quantity (i.e. the concentration of a substance in the air under investigation that is not the measurand), or an external influence quantity (i.e. a quantity that is not the measurand nor the concentration of a substance in the air mass under investigation)

NOTE Examples are:

- presence of interfering gases in the flue gas matrix (interferent influence quantity);
- temperature of the surrounding air (external influence quantity);
- atmospheric pressure (external influence quantity);
- pressure of the gas sample (external influence quantity).

3.14
interference
response of the analyser to interferents

3.15
interferent
component of the air sample, excluding the measured constituent, that effects the output signal

3.16
lag time
time interval from the instant at which a step change of sample concentration occurs at the inlet of the analyser to the instant at which the output reading reaches a level corresponding to 10 % of the stable output reading

3.17
lag time (fall)
lag time for a negative step change

3.18
lag time (rise)
lag time for a positive step change

3.19
limit value
level fixed on the basis of scientific knowledge, with the aim of avoiding, preventing or reducing harmful effects on human health and/or the environment as a whole, to be attained within a given period and not to be exceeded once attained

3.20
lack of fit
maximum deviation of the average of a series of measurements at the same concentration from the linear regression line

3.21
long-term drift
difference between zero or span readings over a determined period of time (e.g. period of unattended operation)

3.22
monitoring station
enclosure located in the field in which a CO analyser has been installed in such a way that its performance and operation comply with the prescribed requirements

3.23
parallel measurement
two measurements from different analysers, sampling from one and the same sampling manifold, starting at the same time and ending at the same time

3.24
performance characteristic
one of the parameters assigned to equipment in order to define its performance

3.25
performance criterion
limiting quantitative numerical value assigned to a performance characteristic, to which conformance is tested

3.26
period of unattended operation
time period over which the drift is within the performance criterion for long-term drift

3.27
repeatability (of results of measurement)
closeness of the agreement between results of successive individual measurements of the same measurand carried out under the same conditions of measurement [1]

NOTE These conditions are called laboratory repeatability conditions and include:

— the same measurement procedure;
— the same observer;
— the same analyser, used under the same conditions;
— at the same location;
— repetition over a short period of time.

3.28
reproducibility under field conditions
closeness of the agreement between the results of simultaneous measurements with two analysers in ambient air carried out under the same conditions of measurement

NOTE 1 These conditions are called laboratory repeatability conditions and include:

— the same measurement procedure;
— two identical analysers, used under the same conditions;
— at the same monitoring station;
— the period of unattended operation.

NOTE 2 In this document the reproducibility under field conditions is expressed as a value with a level of confidence of 95 %.

3.29
response time
time interval from the instant at which a step change of sample concentration occurs at the inlet of the analyser to the instant at which the output reading reaches a level corresponding to 90 % of the stable output reading

3.30
response time (fall)
response time at a negative step change

NOTE Response time (fall) is the sum of the lag time (fall) and the fall time.

3.31
response time (rise)
response time at a positive step change

NOTE Response time (rise) is the sum of the lag time (rise) and the rise time.

3.32
rise time
difference between the response time (rise) and the lag time (rise)

3.33
sampled air
ambient air that has been sampled through the sampling inlet and sampling system

3.34
sampling inlet
entrance to the sampling system where ambient air is collected from the atmosphere

3.35
short-term drift
difference between zero or span readings at the beginning and end of a 12 h period

3.36
standard uncertainty
uncertainty of the result of a measurement expressed as a standard deviation

[ENV 13005]

3.37
surrounding temperature
temperature of the air directly surrounding the analyser (temperature inside the monitoring station or laboratory)

3.38
type approval
decision taken by a designated body that the pattern of an analyser conforms to the requirements as laid down in this document

3.39
type approval test
examination of two or more analysers of the same pattern which are submitted by a manufacturer to a designated body including the tests necessary for approval of the pattern

4 Symbols and abbreviated terms

For the purposes of this document, the following symbols and abbreviated terms apply

A_a	availability of the analyser;
av	average concentration of the measurand during the field test;
b_{gp}	sensitivity coefficient of the analyser to sample gas pressure change expressed as a percentage of the measured value, obtained during the laboratory type approval test;
b_{gt}	sensitivity coefficient of the analyser to sample gas temperature change;
b_{st}	sensitivity coefficient of the analyser to surrounding air temperature change;
b_V	sensitivity coefficient of the analyser to electrical voltage change;
C	CO concentration of the applied gas;
C_{P1}	average concentration of the measurements at sampling gas pressure P_1;
C_{P2}	average concentration of the measurements at sampling gas pressure P_2;
C_R	concentration of the reference standard;
C_s	concentration of site standard;
$C_{s,1}$	average concentration of the measurements at span level at the beginning of the drift period;
$C_{s,2}$	average concentration of the measurements at span level at the end of the drift period;
c_t	test gas concentration;
C_{T1}	average concentration of the measurements at sample gas temperature T_1;
C_{T2}	average concentration of the measurements at sample gas temperature T_2;
C_{V1}	average concentration reading of the measurements at voltage V_1;
C_{V2}	average concentration reading of the measurements at voltage V_2;
$C_{z,1}$	average concentration of the measurements at zero at the beginning of the drift period;
$C_{z,2}$	average concentration of the measurements at zero at the end of the drift period;
$C_{\mathrm{const}}^{\mathrm{av}}$	average of at least four independent measurements during the constant concentration period (t_c);
$C_{\mathrm{var}}^{\mathrm{av}}$	average of at least four independent measurements during the variable concentration period (t_v);
$\overline{d_\mathrm{f}}$	average difference of parallel measurements;
$d_{f,i}$	the ith difference in a parallel measurement;
$D_{l,s}$	long-term drift at span concentration c_t;
$D_{l,z}$	long-term drift at zero;

$D_{s,s}$	short-term drift at span level;
$D_{s,z}$	short-term drift at zero;
D_{sc}	difference sample/calibration port;
E_{ss}	sample system collection efficiency;
F_r	response factor in concentration units per voltage output of the analyser;
P_1	sampling gas pressure P_1;
P_2	sampling gas pressure P_2;
R_d	mean analyser response to the test gas directly sampled by the analyser;
R_m	mean analyser response to the test gas via the sample manifold;
$s_{r,z}$	repeatability standard deviation at zero;
$s_{r,ct}$	repeatability standard deviation at concentration c_t;
$s_{r,f}$	reproducibility standard deviation under field conditions is the repeatability standard deviation;
s_l	repeatability standard deviation;
T	surrounding air temperature;
T_1	sample gas temperature T_1;
T_2	sample gas temperature T_2;
t_d	relative difference between response time (rise) and response time fall;
t_f	response time (fall);
T_l	surrounding air temperature at the laboratory;
$t_{n-1,\ 0,05}$	two-sided Students t-factor at a confidence level of 0,05, with n-1 degrees of freedom;
t_r	response time (rise);
t_t	time period of the field test minus the time for calibration, conditioning and maintenance;
t_u	total time period with validated measuring data;
t_V	whole number of t_{S02} and t_{zero} pairs;
V_1	minimum voltage V_{min} (V) specified by the manufacturer;
V_2	maximum voltage V_{max} (V) specified by the manufacturer;
V_r	voltage obtained when the reference standard is injected;
V_s	voltage obtained when the site standard is injected;
V_z	voltage obtained when zero gas is injected;
$\overline{x}$	average of measurements;

x_1 first average of the measurements at T_l just after calibration;

x_2 second average of the measurements at T_l just before calibration;

X_z difference between the readings of the recent zero check and the most recent calibration;

x_{ct} average of the measurements at concentration c_t;

X_{av} averaging effect;

x_c average of the measurements using the calibration port;

$X_{CO2,z,ct}$ influence quantity of CO_2 with concentration 500 µmol/mol;

$X_{H2O,z,ct}$ influence quantity of H_2O with concentration 19 mmol/mol;

x_i the *i*th measurement;

$X_{int,ct}$ influence quantity of the interferent at concentration c_t;

$X_{int,z}$ influence quantity of the interferent at zero;

X_l lack of fit (largest residual from the linear regression function);

$X_{NO,z,ct}$ influence quantity of NO with concentration 1 µmol/mol;

$X_{N2O,z,ct}$ influence quantity of N_2O with concentration 50 nmol/mol;

x_s average of the measurements using the sample port;

X_s difference between the readings of two consecutive span checks;

x_T average of the measurements at T_{min} or T_{max};

X_z difference between the readings of two consecutive zero checks;

$(x_{1,f})_i$ the *i*th measurement result of analyser 1;

$(x_{2,f})_i$ the *i*th measurement result of analyser 2 at the same time as the measurement of analyser 1;

Z_1 reading of the first zero check;

Z_2 reading of the second zero check;

ΔP_m measured pressure drop induced by the manifold pump;

ΔR_a change in the analyser's response due to the influence of the pressure drop induced by the manifold pump, expressed as a percentage;

5 Principle

5.1 General

This document describes the method for the measurement of the concentration of carbon monoxide in ambient air by means of the non-dispersive infrared spectrometry principle. The requirements, the specific components of the non-dispersive infrared analyser and its sampling system are described. For the analyser a number of performance characteristics with associated minimum performance criteria are given. The actual values of these performance characteristics for a specific type of analyser have to be determined in a so-called type approval test for which procedures have been described. The type approval test comprises a laboratory and a field test. The selection of a suitable analyser for a specific measuring task in the field is based on the calculation of the expanded uncertainty of the measuring method. In this expanded uncertainty calculation the actual values of the various performance characteristics of a type approved analyser and the site-specific conditions at the monitoring station are taken into account. The expanded uncertainty of the method shall meet the requirements of the (EU) legislation. Requirements and recommendations for quality assurance and quality control are given for the measurements in the field (see 9.4).

5.2 Measuring principle

Ambient CO concentration is measured with use of non-dispersive infrared methods. The attenuation of infrared light passing through a sample cell is a measure of the concentration of CO in the cell, according to the Lambert-Beer law. Not only CO but also most heteroatomic molecules will absorb infrared light, in particular water and CO_2 have broad bands that can interfere with the measurement of CO. Different technical solutions have been developed to suppress cross-sensitivity, instability and drift in order to design continuous monitoring systems with acceptable properties. For instance:

— measuring IR absorption of a specific wavelength (4,7 µm for CO);

— dual-cell monitors, using a reference cell filled with clean air (compensation for drift);

— gas filter correlation, “measuring” over a range of wavelengths.

Special attention has to be paid to infrared radiation absorbing gases such as water vapour, carbon dioxide, nitrous oxide and hydrocarbons.

The concentration of carbon monoxide is measured in volume/volume units (if the analyser is calibrated using a volume/volume standard). The final results for reporting are expressed in mg/m^3 using standard conversion factors (see Clause 10).

5.3 Type approval test

The type approval test is based on the evaluation of performance characteristics determined under a prescribed series of tests. In this document test procedures are described for the determination of the actual values of the performance characteristics for at least one analyser in a laboratory and two analysers in the field. These tests shall be performed by a designated body. The evaluation for type approval of an analyser is based on the calculation of the expanded uncertainty in the measurement result derived from the numerical values of the tested performance characteristics and compared with a prescribed maximum uncertainty.

The type approval of an analyser and subsequent OA and QC procedures provide evidence that the defined requirements concerning data quality laid out in relevant EU directives can be satisfied. Appropriate experimental evidence shall be provided by:

— type approval tests performed under conditions of intended use of the specified method of measurement, and

— calculation of expanded uncertainty of results of measurement by reference to ENV 13005.

Field operation and quality control:

Prior to the installation and operation of a type approved analyser at a monitoring station, an expanded uncertainty calculation shall be performed with the actual values of the performance characteristics, obtained during the type approval tests, and the site-specific conditions at that monitoring station. This calculation shall be used to demonstrate the suitability of a type-approved analyser under the actual conditions present at that specific monitoring station.

After the installation of the approved analyser at the monitoring station its correct functioning shall be tested.

Requirements for quality assurance and quality control are given for the operation and maintenance of the sampling system, as well as for the analyser, to ensure that the uncertainty of subsequent measurement results obtained in the field is not compromised.

5.4 Field operation and quality control

Prior to the installation and operation of a type approved analyser at a monitoring station, an expanded uncertainty calculation shall be performed with the actual values of the performance characteristics, obtained during the type approval tests, and the site-specific conditions at that monitoring station. This calculation shall be used to demonstrate the suitability of a type-approved analyser under the actual conditions present at that specific monitoring station.

After the installation of the approved analyser at the monitoring station its correct functioning shall be tested.

Requirements for quality assurance and quality control are given for the operation and maintenance of the sampling system, as well as for the analyser, to ensure that the uncertainty of subsequent measurement results obtained in the field is not compromised.

6 Sampling equipment

6.1 General

Depending on the installation of the non-dispersive infrared analyser at a monitoring station, a single sampling inlet for the analyser may be chosen. Alternatively sampling can take place from a common sampling inlet with a sampling manifold to which other analysers and equipment may be attached. Conditions and layout of the sampling equipment will contribute to the expanded uncertainty of the measurement; to minimise this contribution to the expanded uncertainty, performance criteria for the sampling equipment are given in the following subclauses.

NOTE In Annex B different arrangements of the sampling equipment are schematically presented.

6.2 Sampling location

The location where the ambient air shall be sampled and analysed is not specified, as this depends on the measurement requirements (such as measurements in e.g. a rural area or urban background area).

NOTE For guidance on sampling points on a micro-scale, see Annex C.

6.3 Sample inlet and sampling line

A specific method for sampling or sampling equipment is not described. It is accepted that the equipment shall fulfil the following minimum requirements.

The sample inlet shall be constructed in such a way that ingress of rainwater into the sampling line (or system) is prevented. The sampling line shall be as short as practicable to minimise the residence time.

The material of the sample inlet as well as the sampling line (or system) may influence the composition of the sample. Materials such as polytetrafluoroethylene (PTFE), perfluoro-ethylene-propylene (FEP), borosilicate glass or stainless steel shall be used. The influence of the sampling system on the measured concentrations of carbon monoxide due to losses shall be less than 2 %.

NOTE The sample line may be moderately heated to avoid condensation. Condensation may occur in the case of high ambient temperature and/or humidity.

The influence of the pressure drop due to the sampling line including the particulate filter shall be such that it causes less than 1 % of the output signal of the analyser.

In the case where a sampling manifold is used, an additional pump is necessary with sufficient capacity to meet the sampling requirements stated in the previous subclauses (see also Annex B).

Depending on the location of the particulate filter in the sampling system (see 6.4), the sampling line may be contaminated by deposition of dust. This may induce losses of CO in the sampling line. The sampling system shall be cleaned (as stated in 9.4.2) with a frequency which is dependent on the site-specific conditions.

The sampling line, manifold and filter shall be conditioned (at initial installation and after each cleaning) to avoid temporary decrease in the measured CO concentrations. Both sampling line and filter shall be conditioned with ambient air for a period at least 30 min at the nominal sample flow rate. The measured concentration during these periods shall not be included in the calculation of the hourly and annual averages.

These conditioning periods shall not be included in the calculation of the availability of the analyser during the type approval test (8.5.7) and during field operation (11.2.3). Conditioning may also be done in the laboratory before installing, in the sampling system.

6.4 Particulate filter

A particulate filter shall be placed in the sample line before the inlet of the analyser. This particulate filter shall retain all particles likely to alter the performance of the analyser. It shall be made of PTFE. The material of the filter housing shall be chemically inert to CO.

NOTE 1 A pore size of the filter of 5 µm usually fulfils this requirement.

NOTE 2 Suitable materials for the filter housing are for example PTFE, PFA, FEP and borosilicate glass.

The particulate filter shall be changed periodically depending on the dust loading at the sampling site (as indicated in 9.4.2). The filter housing shall be cleaned at least every six months. Overloading of the particulate filter may cause loss of carbon monoxide by absorption on the particulate matter and can increase the pressure drop in the sampling line.

6.5 Control and regulation of sample flow rate

The sample flow rate into the analyser shall be maintained within the specifications of the manufacturer of the analyser.

6.6 Sampling pump for the manifold

When a sampling manifold is used, a pump (or fan) is necessary for sampling ambient air and drawing of the sampled air through the sampling manifold. The inlet of the sampling pump (or fan) for the sampling manifold shall be located at the end of the sampling manifold (see Annex B). The sampling pump or fan shall have sufficient rating to ensure that all analysers connected to the manifold are supplied with the required amount of air and to ensure that the residence time of a sample from inlet until entering the analyser is less than 5 s. To verify functioning of this pump, it is recommended that a flow alarm system is installed. An example of a sampling manifold is given in Annex B.

The influence of the pressure drop induced by the manifold sampling pump on the measured concentration shall be < 1 %.

7 Analyser equipment

7.1 General

In Annex D schematic diagrams are given of two types of non-dispersive infrared analysers.

The non-dispersive infrared analyser consists of the principal components that are described in 7.3 to 7.6.

7.2 Interferences due to infrared absorbing gases

7.2.1 General

As various gases absorb infrared radiation, interference from these gases can occur when their infrared absorption bands coincide or overlap the CO infrared absorption bands. The degree of interference varies among individual NDIR analysers. In general, gas correlation spectrophotometers are less sensitive to the influence of interferences (see 5.2).

7.2.2 Water vapour

The primary interferent is water vapour.

NOTE Water vapour interference can be minimised by using one or more of the following measures:

- drying the sampled air with use of a semi permeable membrane or a similar drying agent;
- maintaining a constant humidity in the sample and calibration gases by refrigeration or saturation and making a volume correction for the constant humidity;
- using narrow-band optical filters.

7.2.3 Carbon dioxide

Interference may be caused by carbon dioxide (CO_2).

NOTE The effect of CO_2 interference at concentrations normally present in ambient air is minimal; that is, 340 µmol/mol volume fraction may give a response equivalent to 0,2 µmol/mol volume fraction [3]. If necessary, CO_2 may be scrubbed with soda lime.

7.2.4 Hydrocarbons

Hydrocarbons at concentrations normally found in the ambient air do not ordinarily interfere.

NOTE 500 µmol/mol volume fraction of methane may give a response equivalent to 0,5 µmol/mol volume fraction [3].

7.3 Details about analyser equipment

The main parts of the analyser consist of an infrared source, one or two absorption cells, an infrared detector and a chopper wheel for modulation of the infrared radiation to the detector.

In general the infrared source consists of a heated coil combined with optics for dividing the IR beam into two beams of equal intensity in the case of a dual-cell analyser.

As an example, the infrared detector can be a pneumatic radiation detector that selectively registers radiation from the sample cell and alternately (in the case of a dual-cell analyser). This detector consists of a cell filled with a high concentration of CO connected with a compensating cell (see 5.2).

The infrared source, the absorption cells and detector shall be mounted vibration-free.

The materials used in the analyser shall be inert to corrosion.

7.4 Pressure measurement

The output signal of the analyser is proportional to the density of CO (number of CO molecules) present in the absorption cell and depends on the pressure in the absorption cell. Variations of internal pressure shall be measured and the signal corrected.

7.5 Flow rate indicator

A flow rate indicator shall be included in the analyser.

7.6 Sampling pump for the analyser

The sampling pump is situated at the outlet of the analyser, and draws the sample through the analyser. It can be separate or part of the analyser. In any case it shall be capable of operating within the specified flow requirements of the manufacturer of the analyser and pressure conditions required for the NDIR absorption cell.

8 Type approval of CO-NDIR analysers

8.1 General

The determination of the concentration of ozone in ambient air has to fulfil the requirement of a maximum uncertainty in the measured values, which is prescribed by EU legislation. In order to achieve an uncertainty less than (or equal to) this required uncertainty, the analyser has to fulfil all the criteria for a number of performance characteristics, which are given in this document. The values of the selected performance characteristics shall be evaluated by means of laboratory and field tests. By combining the values of the selected performance characteristics in an uncertainty calculation, a judgement can be made whether or not the analyser meets the criterion of maximum uncertainty prescribed by EU legislation.

This process of assessment (type approval test) of the values of the performance characteristics comprises laboratory and field tests and the calculation of the expanded uncertainty. At least two analysers shall be tested in the laboratory and the same analysers shall be tested during the field test. All tested analysers are required to pass.

A designated body shall perform the type approval tests. The type approval shall be awarded by or on behalf of the competent authority.

NOTE It is recommended that the designated body for the type approval test is accredited for these activities according to EN ISO/IEC 17025.

8.2 Relevant performance characteristics and performance criteria

The performance characteristics, which shall be determined during a laboratory and field test, and their related performance criteria, are given in Table 1.

A designated body shall perform the determination of the performance characteristics stated in Table 1 during the laboratory test and field test according to the procedures described in 8.4 and 8.5.

Table 1 — Relevant performance characteristics and criteria

No.	Performance characteristic	Symbol	Clause	Lab. test	Field test	Performance criterion for CO
1	Repeatability standard deviation at zero	$s_{r,z}$	8.4.5	x		≤ 1,0 µmol/mol
2	Repeatability standard deviation at concentration c_t (at a level of the 8-hour mean limit value)	$s_{r,ct}$	8.4.5	x		≤ 3,0 µmol/mol
3	Lack of fit (residual from the linear regression function)		8.4.6			
3a	Largest residual from the linear regression function at concentrations higher than zero	X_l		x		≤ 4,0 % of the measured value [b]
3b	Residual at zero	$X_{l,z}$		x		≤ 0,20 µmol/mol
4	Sensitivity coefficient of sample gas pressure	b_{gp}	8.4.7	x		≤ 0,70 µmol/mol/kPa
5	Sensitivity coefficient of sample gas temperature	b_{gt}	8.4.8	x		≤ 0,30 µmol/mol/K
6	Sensitivity coefficient of surrounding temperature	b_{st}	8.4.9	x		≤ 0,30 µmol/mol/K
7	Sensitivity coefficient of electrical voltage	b_V	8.4.10	x		≤ 0,30 µmol/mol/V
8	Interferents at zero and at concentration c_t (at a level of the 8-hour mean limit value) [a]		8.4.11			
8a	H_2O with concentration 19 mmol/mol [b]	$X_{H2O,z,ct}$		x		≤ 1,0 µmol/mol
8b	CO_2 with concentration 500 µmol/mol	$X_{CO2,z,,ct}$		x		≤ 0,5 µmol/mol
8c	NO with concentration 1 µmol/mol	$X_{NO,z,ct}$		x		≤ 0,5 µmol/mol
8d	N_2O with concentration 50 nmol/mol	$X_{N2O,z,ct}$		x		≤ 0,5 µmol/mol
9	Averaging effect	X_{av}	8.4.12	x		≤ 7,0 % of the measured value
10	Reproducibility standard deviation under field conditions	$s_{r,f}$	8.5.5		x	≤ 5,0 % of the average of a three month period
11	Long-term drift at zero	$D_{l,z}$	8.5.4		x	≤ 0,50 µmol/mol
12	Long-term drift at span level [c]	$D_{l,s}$	8.5.4		x	≤ 5,0 % of maximum of certification range

No.	Performance characteristic	Symbol	Clause	Lab. test	Field test	Performance criterion for CO	
13	Short-term drift at zero	$D_{s,z}$	8.4.4	x		≤ 0,10	µmol/mol over 12 h
14	Short-term drift at span level [c]	$D_{s,s}$	8.4.4	x		≤ 0,60	µmol/mol over 12 h
15	Response time (rise)	t_r	8.4.3	x		≤ 180	s
16	Response time (fall)	t_f	8.4.3	x		≤ 180	s
17	Difference between rise time and fall time	t_d	8.4.3	x		≤ 10	% relative difference, or 10 s, whatever is the greatest
18	Difference between sample/calibration port [d]	D_{sc}	8.4.13	x		≤ 1,0	%
19	Period of unattended operation		8.5.6		X	3	months or less if manufacturer indicates a shorter period, but not less than 2 weeks
20	Availability of the analyser	A_a	8.5.7		X	> 90	%

[a] performance criterion is set both at zero and span level
[b] a H_2O-concentration of 19 mmol/mol is about 80 % RH at 293 K and 101,3 kPa
[c] span level is 70 % to 80 % of the certification range
[d] if relevant

NOTE µmol/mol = ppm; nmol/mol = ppb

8.3 Design changes

The manufacturer shall report any (design) change of the previous type approved monitor to the compenent authority. The underlying idea of the (design) change, as well as the expected consequences with respect to the performance characteristics as stated in Table 1, shall be reported by the manufacturer. The responsible authority shall decide whether or not new tests are necessary. Excluded from reporting are changes in:

— material of the instrument frame (housing);

— electronic connectors;

— bolts, nuts, cable binders and other fixing materials.

8.4 Procedures for determination of the performance characteristics during the laboratory test

8.4.1 General

A designated body shall perform the determination of the performance characteristics in the laboratory as a part of the type approval test. The quality of the materials and equipment used in the described test procedures shall be in accordance with the requirements given in this document. The tests shall be performed on at least two analysers in the laboratory test.

8.4.2 Test conditions

8.4.2.1 General

Before operating the analyser, the operating instructions of the manufacturer shall be followed particularly with regard to the set-up of equipment and the quality and quantity of the consumable products necessary.

The analyser should be allowed to warm up during the time specified by the manufacturer before undertaking any tests. If the warm-up time is not specified, a minimum of 4 h is recommended.

When applying test gases to the analyser, the test gas system shall be operated sufficiently long before starting the tests in order to stabilise the concentrations applied to the analyser.

Most analyser systems are able to give an output signal as a moving average over an adjustable period of time and some systems automatically change this integration time as a function of the frequency of the fluctuations in concentration of the detected pollutant. These options are typically used in order to smooth output data. It needs to be demonstrated that the set value for the averaging time or the use of an active filter will not influence the result of the averaging test and response time test.

During laboratory and field tests for the type approval the settings of the monitor shall be as the manufacturer requires. All settings shall be noted down in the test report.

8.4.2.2 Parameters

During the test for each individual performance characteristic, the values of the following parameters shall be stable within the specified range given in Table 2.

Table 2 — Set points and stability of test parameters

Parameter	Set points	Stability
Sample gas pressure	Manufacturer's specification (except for the sample gas pressure test in, see 8.4.7)	± 0,2 kPa
Sample gas temperature	20 °C to 23 °C (except for the sample gas temperature test, see 8.4.8)	± 2 °C
Surrounding air temperature	20 °C to 23 °C (except for the temperature of the surrounding air test, see 8.4.9)	± 2 °C
Electrical voltage [a]	At nominal line voltage and within manufacturer's specifications (except for the voltage dependence test, see 8.4.10)	± 1 %
Sample flow to the analyser	Manufacturer's specification	± 1 %
[a] For an analyser operating on direct current the type approval test of voltage variation shall be carried out over the range of ±10 % of the nominal voltage.		

8.4.2.3 Test gases

For the determination of the various performance characteristics, test gases ((synthetic) air containing a certain CO concentration) traceable to national standards shall be used, unless otherwise stated in this document. Various methods for the generation of test gases are given in Table 3.

Table 3 — Methods for preparation of test gases

Method	Description	Standard to be used
Commercial cylinder	Gas cylinder (CO gas mixture in (synthetic) air)	ISO 6143
Static dilution	Preparation by means of injecting known amounts of CO in a	ISO 6144

Method	Description	Standard to be used
	known volume	
Dynamic dilution	Dynamic blending of cylinder gas with (synthetic) air, zero air	ISO 6145

The uncertainties in zero and span gases used for the laboratory and field tests shall be proven to be insignificant. Possible contamination of zero and span gas shall not significantly influence the results of the laboratory tests. Therefore the span gases and zero gas shall meet the following specifications:

Maximum permitted uncertainties in the concentration of gases used for laboratory tests: ± 3 %.

Specification of the purity of test gases and zero gas (expressed as absolute value) is given in Table 4.

Table 4a — Specification for purity of span gas

Pollutant	Concentration
CO_2	< 4 µmol/mol
NO	< 1 nmol/mol
N_2O	< 0,5 nmol/mol
water	< 150 µmol/mol

Table 4a — Specification for purity of zero gas

Pollutant	Concentration
CO_2	< 4 µmol/mol
NO	< 1 nmol/mol
N_2O	< 0,5 nmol/mol
water	< 150 µmol/mol
CO	< 0,1 µmol/mol

NOTE It is advised to perform the field tests with the same set of cylinders and zero air generators, exclusively reserved for the tests. The stability of both the zero air and the span gas should be guaranteed over a period longer than the test period.

8.4.3 Response time

8.4.3.1 General requirements

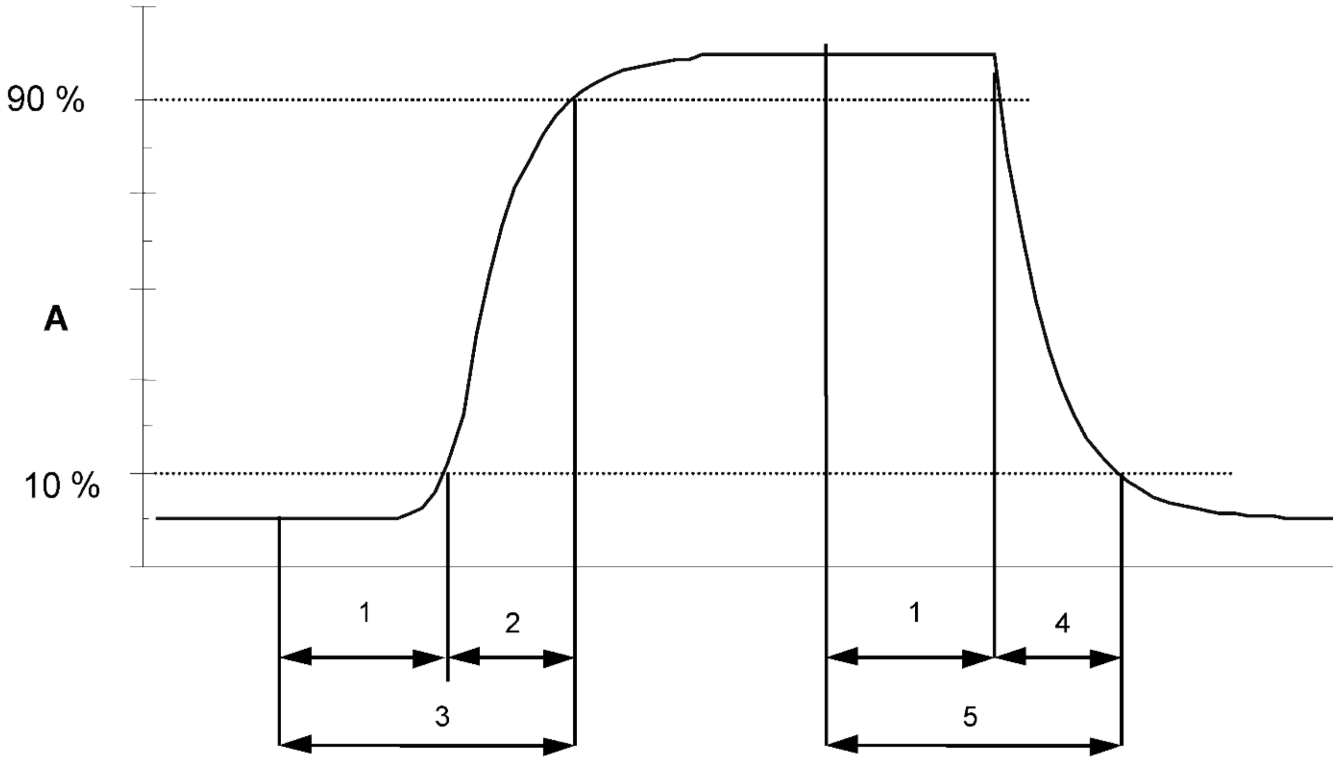
The response time of the analyser shall be determined at the nominal sample flow rate specified by the manufacturer.

The sample flow rate shall be kept constant within the requirements as given in 8.4.2 (± 1 %) during the test.

8.4.3.2 Test procedure

The determination of the response time shall be carried out by applying to the analyser a step function in the concentration from less than 20 % to about 80 % of the maximum of the certification range and vice versa (see Figure 1).

The change from zero gas to span gas needs to be made almost instantly with the use of a suitable valve. The valve outlet shall be mounted direct to the inlet of the analyser, and both zero gas and span gas shall have the same “oversupply“ which is vented with the use of a tee. The gas flows of both zero gas and span gas shall be chosen in such a way that the dead time in the valve and tee can be neglected compared to the lag time of the analyser system. The step change is made by switching the valve from zero gas to span gas. This event needs to be timed and is the start (t = 0) of the (rise) lag time according to Figure 1. When the reading of 98 % of the applied concentration has been reached, the span gas can be changed to zero again, and this event is the start (t = 0) of the (fall) lag time. When the reading of 2 % of the applied concentration has been reached the whole cycle as shown in Figure 1 is complete.



Key

A Analyser response

1 Lag time

2 Rise time

3 Response time (rise)

4 Fall time

5 Response time (fall)

Figure 1 — Diagram illustrating the response time

The relative difference in response times shall be calculated according to:

$$t_\mathrm{d} = \left| \frac{t_\mathrm{r} - t_\mathrm{f}}{t_\mathrm{r}} \right| \times 100\% \qquad (1)$$

where

t_d is the relative difference between response time (rise) and response time (fall) (%);

t_r is the average response time (rise) (average of 4 measurements) (s);

t_f is the average response time (fall) (average of 4 measurements) (s).

t_r, t_f and t_d shall comply with the performance criterion in Table 1.

8.4.4 Short-term drift

After the required stabilisation period (8.4.2.1), the analyser shall be adjusted at zero and span level (around 70 % to 80 % of the maximum of the certification range). Wait the time equivalent to one independent reading and then record 20 individual measurements first at zero and then at span concentration. From these 20 measurements the average is calculated for zero and span level.

The analyser shall be kept running under the laboratory conditions (8.4.2.2) whilst analysing ambient air. After a period of 12-hours zero and span gas is fed to the analyser. Wait the time equivalent to one independent reading and then record 20 individual measurements first at zero and than at span concentration. The averages for zero and span level shall be calculated.

NOTE Compliance with this test shows that drift is not the dominant factor in any of the test results.

The short-term drift at zero and span level shall be calculated as follows:

$$D_{s,z} = (C_{z,2} - C_{z,1}) \tag{2}$$

where

$D_{s,z}$ is the 12-hour drift at zero (µmol/mol);

$C_{z,1}$ is the average of the zero gas measurements at the beginning of the drift period (just after calibration) (µmol/mol);

$C_{z,2}$ is the average of the zero gas measurements at the end of the drift period (12 h) (µmol/mol).

$D_{s,z}$ shall comply with the performance criterion in Table 1.

$$D_{s,s} = (C_{s,2} - C_{s,1}) - D_{s,z} \tag{3}$$

where

$D_{s,s}$ is the 12-hour drift at span (µmol/mol);

$C_{s,1}$ is the average of the span gas measurements at the beginning of the drift period (just after calibration) (µmol/mol);

$C_{s,2}$ is the average of the span gas measurements at the end of the drift period (12 h) (µmol/mol).

$D_{s,s}$ shall comply with the performance criterion in Table 1.

8.4.5 Repeatability standard deviation

After waiting the time equivalent of one independent reading 20 individual measurements both at zero concentration and at a test concentration (c_t) similar to the 8-hour mean limit value shall be performed. From these measurements the repeatability standard deviation (s_r) at zero concentration and at concentration c_t shall be calculated according to:

$$s_i = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} \tag{4}$$

where

s_r is the repeatability standard deviation (µmol/mol);

x_i is the *i*th measurement (µmol/mol);

$\overline{x}$ is the average of the 20 measurements (µmol/mol);

n is the number of measurements, n = 20.

The repeatability standard deviation shall be calculated separately for both series of measurements (zero gas and concentration c_t).

s_r shall comply with the performance criterion in Table 1, both at zero and at the test concentration ct (8-hour mean limit value).

8.4.6 Lack of fit

The lack of fit of the analyser shall be tested over the range between 0 % to 95 % of the maximum of the certification range using at least six concentrations (including the zero point). The analyser shall be adjusted at a concentration of about 90 % of the maximum of the certification range. At each concentration (including zero) at least five individual readings shall be performed.

The concentrations shall be applied in the following sequence: 80 %, 40 %, 0 %, 60 %, 20 % and 95 %. After each change in concentration at least four response times shall be taken into account before the next measurement is performed.

The uncertainty in the dilution ratios for the applied concentrations shall be less than 0,5 % with respect to each other.

NOTE The design of the test (number of concentrations and number of repetitions) is such that the deviation from a linear function can be determined with sufficient accuracy. The test is also sufficiently robust to detect the case of non-linearity in the range from zero to some concentration at the lower end of the range as well as non-linearity from that lower end of the range to the higher end of the range.

Calculation of the linear regression function and residuals shall be performed according to Annex A. All the (relative) residuals from the linear regression function shall fulfil the criteria as stated in Table 1.

The largest value of the relative residuals is reported as X_i and shall be taken into account in demonstrating compliance with type approval requirement 1. The value of the relative residual value at the level of the 8-hour mean limit value shall be taken in the calculation of type approval requirement 2 and 4.

8.4.7 Sensitivity coefficient to sample gas pressure

Measurements are taken at a concentration of about 70 % to 80 % of the maximum of the certification range at an absolute pressure of 80 kPa ± 0,2 kPa and at an absolute pressure of 110 kPa ± 0,2 kPa. At each pressure after waiting the time equivalent to one independent reading three individual measurements are recorded. From these 3 measurements the averages at each pressure are calculated.

Measurements at different pressures shall be separated by at least four response times.

The sample gas pressure influence is calculated by:

$$b_{gp} = \left| \frac{(C_{P_1} - C_{P_2})}{(P_2 - P_1)} \right| \tag{5}$$

where

b_{gp} is the sample gas pressure influence (µmol/mol/kPa);

C_{p_1} is the average concentration of the measurements at sampling gas pressure P_1 (µmol/mol);

C_{p_2} is the average concentration of the measurements at sampling gas pressure P_2 (µmol/mol);

P_1 is the sampling gas pressure P_1 (kPa);

P_2 is the sampling gas pressure P_2 (kPa).

b_{gp} shall comply with the performance criterion in Table 1.

8.4.8 Sensitivity coefficient to sample gas temperature

For the determination of the dependence of the sample gas temperature measurements shall be performed at sample gas temperatures of T_1 = 0 °C and T_2 = 30°C. The temperature dependence shall be determined at a concentration of about 70 % to 80 % of the maximum of the certification range. Wait the time equivalent to one independent and record 3 individual measurements at each temperature.

The sample gas temperature, measured at the inlet of the analyser, shall be held constant for at least 30 min.

The influence of sample gas temperature is calculated from:

$$b_{gt} = \frac{\left(C_{T_2} - C_{T_1}\right)}{\left(T_2 - T_1\right)} \tag{6}$$

where

b_{gt} is the sample gas temperature influence (µmol/mol/ °C);

C_{T_1} is the average concentration of the measurements at sample gas temperature T_1 (°C);

C_{T_2} is the average concentration of the measurements at sample gas temperature T_2 (°C);

T_1 is the sample gas temperature T_1 (°C);

T_2 is the sample gas temperature T_2 (°C).

b_{gt} shall comply with the performance criterion in Table 1.

8.4.9 Sensitivity coefficient to the surrounding temperature

The influence of the surrounding temperature shall be determined at the following temperatures (within the specifications of the manufacturer):

1) at the minimum temperature T_{min} = 273 K;

2) at the laboratory temperature (T_l), see 8.4.2.2;

3) at the maximum temperature T_{max} = 303 K.

For these tests a climate chamber is necessary.

The influence shall be determined at a concentration around 70 % to 80 % of the maximum of the certification range. At each temperature setting after waiting the time equivalent to one independent reading three individual measurements at zero and at span shall be recorded.

At each temperature setting, the criteria for warm-up or stabilisation time are to be met according to 8.4.2.1.

The measurements shall be performed in the following sequence of the temperature settings:

T_{lab}, T_{min}, T_{lab} and T_{lab}, T_{max}, T_{lab}

At the first temperature (T_{lab}) the analyser shall be adjusted at zero and at span level (70 % to 80 % of the maximum of the certification range). Then three individual measurements are recorded after waiting the time equivalent to one independent reading at T_l, at T_{min} and again at T_{lab}. This procedure shall be repeated at the temperature sequence of T_{lab}, T_{max}, and at T_{lab}.

In order to exclude any possible drift due to factors other than temperature, the measurements at T_{lab} are averaged, which is taken into account in the following formula for calculation of the surrounding air temperature dependence:

$$b_{st} = \left| \frac{x_T - \frac{x_1 + x_2}{2}}{T - T_l} \right| \tag{7}$$

where

b_{st} is the surrounding air temperature dependence at zero or span and at T_{min} or T_{max} (µmol/mol /°C);

x_T is the average of the measurements at T_{min} or T_{max} (µmol/mol);

x_1 is the first average of the measurements at T_{lab} just after calibration (µmol/mol);

x_2 is the second average of the measurements at T_{lab} just before calibration (µmol/mol);

T_{lab} is the surrounding air temperature at the laboratory (°C);

T is the surrounding air temperature T_{min} or T_{max} (°C).

For reporting the surrounding air temperature dependence the highest value is taken of the two calculations of the temperature dependence at T_{min} and T_{max}.

b_{st} shall comply with the performance criterion in Table 1.

8.4.10 Sensitivity coefficient to electrical voltage

The sensitivity coefficient of electrical voltage shall be determined at both ends of the specified voltage range, V_{min} and V_{max} at zero concentration and at a concentration around 70 % to 80 % of the maximum of the certification range. After waiting the time equivalent to one independent reading three individual measurements at each voltage and concentration level shall be recorded.

The voltage dependence is calculated from:

$$b_v = \frac{(C_{V_2} - C_{V_1})}{(V_2 - V_1)} \tag{8}$$

where

b_v is the voltage influence (µmol/mol/V);

C_{V_1} is the average concentration reading of the measurements at voltage V_1 (µmol/mol);

C_{V_2} is the average concentration reading of the measurements at voltage V_2 (µmol/mol);

V_1 is the minimum voltage V_{min} (V) specified by the manufacturer;

V_2 is the maximum voltage V_{max} (V) specified by the manufacturer.

b_v shall comply with the performance criterion in Table 1.

For an analyser operating on direct current the type approval test of voltage variation shall be carried out over the range of ± 10 % of the nominal voltage.

8.4.11 Interferences

The analyser response to certain interferents, which are expected to be present in ambient air, shall be tested. The interferents can give a positive or negative response. The test shall be performed at zero and at a test concentration (c_t) similar to the 8-hour mean limit value.

The concentration of the mixtures of the test gases with the interferent shall have an uncertainty of less than 5 % and shall be traceable to national standards. The interferents to be tested and their respective concentrations are given in Table 1. The influence of each interferent shall be determined separately. A correction of the concentration of the measurand shall be made for the dilution effect due to addition of an interferent (e.g. water vapour).

After adjustment of the analyser at zero and span level the analyser shall be fed with a mixture of zero gas and the interferent to be investigated with the concentration as given in Table 1. With this mixture one independent measurement followed by two individual measurements shall be carried out. This procedure shall be repeated with a mixture of the measurand at concentration c_t and the interferent to be investigated. The influence quantity at zero and concentration c_t are calculated from:

$$X_{\text{int},z} = x_1 \tag{10}$$

$$X_{\text{int},ct} = x_2 - c_t \tag{11}$$

where

$X_{\text{int,z}}$ is the influence quantity of the interferent at zero (µmol/mol);

x_1 is the average of the measurements at zero (µmol/mol).

$X_{\text{int,ct}}$ is the influence quantity of the interferent at concentration c_t (µmol/mol);

x_2 is the average of the measurements at concentration c_t (µmol/mol);

c_t is the concentration of the applied gas at the level of the 8-hour mean limit value (µmol/mol).

The influence quantity of the interferences shall comply with the performance criteria in Table 1, both at zero and at concentration c_t.

8.4.12 Averaging test

The averaging test gives a measure of the uncertainty in the averaged values caused by short-term concentration variations in the sampled air shorter than the time scale of the measurement process in the analyser. In general the output of an analyser is a result of the determination of a reference concentration (normally zero) and the actual concentration, which takes a certain time.

For the determination of the uncertainty due to the averaging the following concentrations are applied to the analyser and readings are taken at each concentration (see also Figure 2):

— a stepwise varied concentration of CO between zero and concentration c_t (70 % to 80 % of the maximum certification range).

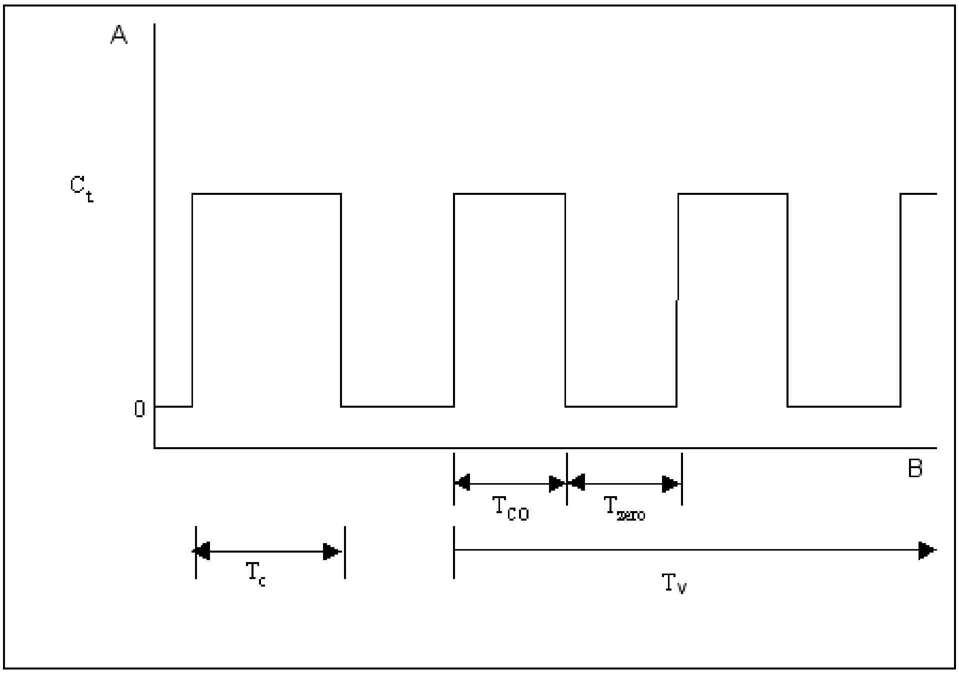
The time period (t_c) of the constant CO concentration shall be at least equal to a period necessary to obtain four independent readings (which equals to at least sixteen response times). The time period (t_v) of the varying CO concentration shall be at least equal to a period to obtain four independent readings. The time period (t_{CO}) for the CO concentration shall be 45 s followed by a period (t_{zero}) of 45 s of zero concentration.

Further:

c_t is the test concentration (µmol/mol);

t_V is a whole number of t_{C0} and t_{zero} pairs, and contains a minimum of three such pairs.

The change from t_{CO} to t_{zero} shall be within 0,5 s. The change from t_C to t_V shall be within one response time of the analyser under test.



Key

A Concentration (µmol/mol)

B Time

Figure 2 — Concentration variations for the averaging effect test

The averaging effect (X_{av}) is calculated according to:

$$X_{av} = \frac{C_{const}^{av} - 2 \times C_{var}^{av}}{C_{const}^{av}} \times 100\% \tag{12}$$

where

X_{av} is the averaging effect (%);

C_{const}^{av} is the average of at least four independent measurements during the constant concentration period (t_c) (µmol/mol);

C_{var}^{av} is the average of at least four independent measurements during the variable concentration period (t_v) (µmol/mol).

8.4.13 Difference sample/calibration port

If the analyser has different ports for feeding sample gas and calibration gas the difference in response of the analyser to feeding through the sample or calibration port shall be tested. The test shall be carried out by feeding the analyser with a test gas with a concentration of 70 % to 80 % of the maximum of the certification range through the sample port. The test shall consist of one independent followed by two individual measurements. After a period of at least 4 response times the test shall be repeated using the calibration port. The difference shall be calculated according to:

$$D_{sc} = \frac{x_s - x_c}{c_t} \times 100\,\% \qquad (13)$$

where

D_{sc} is the difference sample/calibration port (%);

x_s is the average of the measured concentrations using the sample port (µmol/mol);

x_c is the average of the measured concentrations using the calibration port (µmol/mol);

c_t is the concentration of the test gas (µmol/mol).

D_{sc} shall comply with the performance criterion in Table 1.

8.5 Determination of the performance characteristics during the field test

8.5.1 General

The determination of the performance characteristics in the field as a part of the type approval test shall be performed by a designated body. The quality of the materials and equipment used in the described test procedures shall be in accordance with the requirements given in this document.

In the field test during a period of three months 2 analysers are tested for availability (period of unattended operation), reproducibility in the field and long-term drift. The analysers are run in parallel at one and the same sampling point at a selected monitoring station with specific ambient air conditions (8.5.2). Operational requirements are given for the correct determination of the long-term drift and the reproducibility under field conditions (8.5.3).

8.5.2 Selection of a monitoring station for the field test

The selection of a monitoring station is based on the following criteria:

Location:

— traffic orientated station (4 m to 5 m from kerb-side).

Monitoring station facilities:

— sufficient capacity of the sampling manifold;

— enough room to place two analysers with calibration gases and/or calibration facilities;

— surrounding temperature control for the analysers, climate controlled at 20°C ± 4 °C with temperature registration;

— stable electrical voltage.

Other items that could be considered:

— presence of telemetry/telephone facilities for remote surveillance of the functioning of the equipment;

— accessibility.

8.5.3 Operational requirements

After installation of the analysers at the monitoring station the proper functioning of the analysers shall be tested. This comprises (among other things) the proper connections to the sampling manifold, sample gas flows, correct temperatures of e.g. reaction chambers, response to zero and span gases, actual converter efficientcy, data transmission and other items, which shall be judged necessary by the designated body.

After verification of the proper functioning, the analysers shall be adjusted at zero and calibrated at a value of about 80 % of the maximum of the certification range.

During the three-month period, the maintenance requirements by the manufacturer of the analyser shall be followed.

Measurements with zero and span gases shall be performed every two weeks. The concentration c_t of the span gas shall be around 90 % of the maximum of the certification range. One independent followed by 4 individual measurements shall be performed both at zero and at concentration c_t. The measurement results shall be recorded.

To exclude the effect of contamination of the filter when determining the drift of the analyser the zero and span gases shall be fed to the analyser without passing through the filter.

To avoid the possibility that the filter loading affects the results of the intercomparison of the two analysers and to ensure that the filter loading will not compromise the quality of the air pollution data collected, the filter shall be changed just before each bi-weekly calibration. Filters, which had been preconditioned in the laboratory using CO gas mixtures, shall be used.

During the three-month period, no zero and span adjustments shall be made to the analyser, as this will influence the determination of the long-term drift. The measurement data from the analyser shall only be corrected in a mathematical way assuming a linear drift since the last zero and span check.

If an auto rescaling function or self-correction function is included and considered “normal operational condition”, it shall be enabled during the field tests. The magnitude of any self correction shall be available tot the test laboratory. The magnitude of the auto zero and the auto span drift corrections over the period of unattended operation (long-term drift) both have the same restrictions as laid down in the performance characteristics.

For the determination of the various performance characteristics, test gases (air containing a certain CO concentration) shall be used, traceable to national standards, unless otherwise stated in this document. Various methods for the generation of test gases are given in Table 3 (see 8.4.2.3).

The uncertainties in zero and span gases used for the field tests shall proven to be insignificant. Possible contamination of zero and span gas shall not significantly influence the results of the laboratory and field tests. Therefore the test gases and zero gas shall meet the following specifications:

Maximum permitted uncertainties in the concentration of gases used for field tests: ± 5 %;

Specification of the purity of test-gases and zero gas (expressed as absolute value) as given in Table 4.

NOTE It is advised to perform the field tests with the same set of cylinders and zero air generators, exclusively reserved for the tests. The stability of both the zero air and the span gas should be guaranteed over a period longer than the test period.

8.5.4 Long-term drift

After each bi-weekly calibration the drift of the analysers under test shall be calculated at zero and at span following the procedures as given underneath. If the drift compared to the initial calibration exceeds one of the performance criteria for drift at zero or span level, the “period of unattended operation” equals the number of weeks till the observation of the infringement, minus two weeks. For further (uncertainty) calculations the values for “long term drift” are the values for zero and span drift over the period of unattended operation.

At the beginning of the drift period five individual measurements are recorded (after waiting the time equivalent to one independent measurement just after the calibration) at zero and at span level. The long-term drift is calculated as follows:

$$D_{l,z} = (C_{z,2} - C_{z,1}) \tag{14}$$

where

$D_{l,z}$ is the drift at zero (μmol/mol);

$C_{z,1}$ is the average concentration of the measurements at zero at the beginning of the drift period (just after the initial calibration) (μmol/mol);

$C_{z,2}$ is the average concentration of the measurements at zero at the end of the drift period (μmol/mol).

$D_{l,z}$ shall comply with the performance criterion in Table 1.

$$D_{l,s} = \frac{(C_{s,2} - C_{s,1}) - D_{l,z}}{C_{s,1}} \times 100\ \% \tag{15}$$

where

$D_{l,s}$ is the drift at span concentration c_t (μmol/mol);

$C_{s,1}$ is the average concentration of the measurements at span level at the beginning of the drift period (just after the initial calibration) (μmol/mol);

$C_{s,2}$ is the average concentration of the measurements at span level at the end of the drift period (μmol/mol).

$D_{l,s}$ shall comply with the performance criterion in Table 1.

NOTE For the determination of a systematic or random drift a graph with the zero and span gas readings can be useful.

8.5.5 Reproducibility under field conditions

The reproducibility standard deviation under field conditions is calculated from the measured hourly averaged data during the three months period.

The difference d_f for each (*i*th) parallel measurement is calculated from:

$$d_{f,i} = (x_{1,f})_i - (x_{2,f})_i \tag{16}$$

The average difference from these measurements is calculated from:

where

$d_{f,i}$ is the i[th] difference in a parallel measurement (μmol/mol);

$(x_{1,f})_i$ is the *i*th measurement result of analyser 1 (μmol/mol);

$(x_{2,f})_i$ is the *i*th measurement result of analyser 2 at the same time as the measurement of analyser 1, (μmol/mol).

The reproducibility (under field conditions) standard deviation ($s_{r,f}$) is calculated according to:

$$s_{r,f} = \frac{\left(\sqrt{\frac{\sum_{i=1}^{n} d_{f,\ i}^{2}}{2n}}\right)}{av} \text{ x } 100 \qquad (18)$$

where

$s_{r,f}$ is the reproducibility standard deviation under field conditions (%);

n is the number of parallel measurements;

av is the avarage value during the field test (µmol/mol).

The reproducibility standard deviation under field conditions, s_r, shall comply with the performance criterion in Table 1.

8.5.6 Period of unattended operation

The period of unattended operation is the time period within which the drift is within the performance criterion for long-term drift. If the manufacturer specifies a shorter period for maintenance, than this will be taken as the period of unattended operation. If one of the analysers malfunctions during the field test, than the field test shall be restarted to show whether the malfunction was coincidental or bad design.

NOTE A minimum period of unattended operation is normally recommended to be at least two weeks.

8.5.7 Period of availability of the analyser

The correct operation of the analysers shall be checked at least every 14 days. It is recommended to perform this check every day during the first 14 days. These checks consists of plausibility checks on the measured values, as well as when available status signals and other relevant parameters. Time, duration and nature of any malfunctioning shall be logged.

The total time period with useable measuring data is the period during the field test during which valid measuring data of the ambient air concentrations are obtained. In this time period the time needed for calibrations, conditioning of sample lines (6.3) and filters (6.4) and maintenance shall not be included.

The availability of the analyser is calculated as:

$$A_\mathrm{a} = \frac{t_u}{t_t} \times 100\,\% \qquad (19)$$

where

A_a is the availability of the analyser;

t_u is the total time period with validated measuring data;

t_t is the time period of the field test minus the time for calibration, conditioning and maintenance.

t_u and t_t shall be expressed in the same units (e.g. hours).

The availability shall comply the criterion in Table 1.

8.6 Expanded uncertainty calculation for type approval

The type approval of the analyser consists of the following steps:

1) the value of each individual performance characteristic tested in the laboratory shall fulfil the criterion stated in Table 1 (see 8.2);

2) the expanded uncertainty calculated from the standard uncertainties due to the values of the specific performance characteristics obtained in the laboratory tests fulfils the criterion as stated in the Council Directive 2000/69/EC. This criterion is the maximum uncertainty of individual measurements for continuous measurements at the 8-hour mean limit value. The relevant specific performance characteristics and the calculation procedure are given in Annex F;

3) the value of each of the individual performance characteristics tested in the field shall fulfil the criterion stated in Table 1 (see 8.2);

4) the expanded uncertainty calculated from the standard uncertainties due to the values of the specific performance characteristics obtained in the laboratory and field tests fulfils the criterion as stated in the Council Directive 2000/69/EC. This criterion is the maximum uncertainty of individual measurements for continuous measurements at the 8-hour mean limit value. The relevant specific performance characteristics and the calculation procedure are given in Annex F.

The instrument can be typed as approved when all 4 requirements are met.

9 Field operation and ongoing quality control

9.1 General

When a type approved analyser has been chosen for a particular measuring task, then the suitability of this analyser shall be evaluated at a specific measuring location. This shall be performed by means of a suitability evaluation described in 9.2.

The analyser shall be installed at a monitoring station in such a way that normal operation of the analyser is not compromised. This implies that the analyser is sheltered and shielded from dust, rain and snow, direct sun radiation, strong temperature fluctuations etc. An enclosure (container or building) with temperature control or air conditioning usually fulfils these requirements. At some locations voltage stabilisers for the power supply may be considered, when voltage fluctuations are expected.

After installation of the analyser at the measuring station the analyser shall be tested for proper operation. This is described in 9.3 (initial installation).

Subsequently, once the analyser at the specific site has been judged to conform with the EU data quality objectives, quality assurance and quality control procedures for ongoing monitoring of carbon monoxide dioxide concentrations shall be followed (as described in 9.4) in order to ascertain that the measured data comply with the uncertainty requirements as given in the EU legislation.

9.2 Suitability evaluation

9.2.1 General

When a type approved analyser has been chosen, the suitability of this analyser shall be evaluated for the site-specific conditions at the monitoring site. For example, temperature fluctuations might be such that the type approved analyser does not fulfil the uncertainty requirements under these conditions. Consequently it may be necessary to control the temperature of the air directly surrounding the analyser. Carbon monoxide test gases traceable to national standards shall be used for the correct calibration of the analyser.

9.2.2 Analyser for a monitoring station or task

An expanded uncertainty calculation for the type approved analyser shall be made according to EN ISO 14956 in combination with the specific circumstances at the monitoring station or site. For the specific concepts of uncertainty calculations, reference is made to ENV 13005 and to 200/69/EC 98/0333 (COD). The site-specific circumstances, which shall be evaluated, are given in Table 6.

Table 6 — Site-specific conditions to be evaluated

Parameters	**Remarks**
Sample pressure variation	The sample gas pressure variation during a whole period of a year shall be estimated.
Sample gas temperature variation	The sample gas temperature variation during a whole period of a year shall be estimated. Sample gas temperature may be controlled by heating or thermostating.
Surrounding air temperature variation	The temperature fluctuation shall be within the range specified in the type approval test. Temperature may be controlled thermostatically.
Voltage variation [a]	The voltage fluctuation shall be within the range in the type approval test. Voltage fluctuations may be controlled by means of voltage stabilizers.
H_2O concentration range	The H_2O concentration range during a whole period of a year shall be estimated.
CO_2 concentration range	The CO_2 concentration range during a whole period of a year shall be estimated.
NO concentration range	The NO concentration range during a whole period of a year shall be estimated.
N_2O concentration range	The N_2O concentration range during a whole period of a year shall be estimated.
m-xylene concentration range	The m-xylene concentration range during a whole period of a year shall be estimated.
Expanded uncertainty of the calibration gas	The expanded uncertainty of the calibration gas shall be included. This implies the uncertainty of the calibration gas itself as well as the uncertainty of any dilution system (where applicable).
Calibration frequency	The intended calibration frequency shall be used for the calculation of the influence of the drift.

[a] For an analyser operating on direct current the type approval test of voltage variation shall be carried out over the range of ± 10 % of the nominal voltage.

If the site-specific conditions are outside the conditions for which the analyser is type approved, then the analyser shall be retested under these site-specific condtions and a revised type approval will be issued.

The analyser can only be used in the tested configuration.

If the analyser complies with the requirements laid down in EU legislation, then that particular analyser may be installed and used at that monitoring station.

The calculations shall be documented.

9.3 Initial installation

When the analyser and the sampling system have been set up at the monitoring station proper functioning of the analyser and sampling system shall be checked. The results of these checks shall fulfil the requirements and limitations as set out by the manufacturer of the analyser as well as the requirements (such as materials used, residence times and so on) given in this document. The compliance with the requirements of the manufacturer and requirements set out in this document shall be documented.

When the concentrations measured by an analyser at a monitoring station are collected by a data-logger/computer system, then the proper functioning of the data collection shall be checked. When the measured data are transmitted to a central computer system, the transmission process shall be checked as well. Checks shall be performed to an extent ensuring that the actual concentrations measured by the analyser are properly recorded in any data collection system.

Subsequently each time parts of the data registration/transmission process are changed, the proper function of the complete process shall be rechecked.

All checks on the proper function of the data collection/transmission system(s) shall be documented.

During the initial installation a lack of fit check shall be performed according to 8.4.6. It is permitted that this test be carried out in the laboratory before installation at the site or after installation at the site.

9.4 Ongoing quality control

9.4.1 General

Quality control is necessary in order to ensure that the uncertainties of the measured values for carbon monoxide in ambient air are kept within the stated limits during extended continuous monitoring periods in the field. This requires that maintenance, test and calibration procedures shall be followed which are essential for obtaining accurate and traceable air quality data. In this clause, procedures for maintenance, checks and calibration are given. These procedures are regarded as a minimum necessary for maintaining the required quality level.

The quality of the materials and equipment used in the described test procedures shall be in accordance with the requirements given in this document and shall not significantly influence the results of these procedures.

NOTE It is recommended that the designated body who is performing the ongoing quality control procedures is accredited according to EN ISO/IEC 17025.

9.4.2 Frequency of calibrations, checks and maintenance

The checks and calibrations together with their frequency are summarised in Table 7. Criteria are also given for readjustment, calibration or maintenance of the analyser.

Table 7 — Required frequency of calibration, checks and maintenance

Calibration, checks and maintenance	Clause	Frequency	Action criteria [e]
Calibration of the analyser	9.5	At least every three months and after repair	
Certfifcation of test gases	9.6.1	At least every six months	Zero: ≥ detection limit Span: ≥ 5,0 % from last certified value
Zero and span check [a]	9.6.2	At least every two weeks [b]	Zero: ≥ 0,2 µmol/mol Span: ≥ 5,0 % of initial span value
Lack of fit check (to be performed in laboratory or in field)	9.6.3, Annex B	Within 1 year after installation and after repair; further frequency depending	lack of fit > 6,0 % of the measured value

Calibration, checks and maintenance	Clause	Frequency	Action criteria [e]
		on the result of test	
Testing sample manifold - influence of pressure drop induced by the manifold pump - sample collection efficiency	 9.6.4.1 9.6.4.2	 At least every three years	 influence > 1 % of measured value influence > 2 % of the measured value
Change of particulate filters [c] of the sampling system at the sampling inlet and/or at the analyser inlet	6.4, 9.7.1	At least every three months [d]	Response to span gas passing the filter is ≤ 97 %
Test of the sampling lines	6.3	At least every six months [d]	≥ 2 % sample loss
Changing of (if applicable): drying material and other consumables	6.4 9.7.2	At least every six months [d]	As required
Regular maintenance of components of the analyser	9.7.3	As required by manufacturer	As required

[a] Span value: recommended concentration of 70 % to 80 % of the certification range.

[b] Recommended every 23 h or 25 h.

[c] The particulate filter shall be changed periodically depending on the dust loading at the sampling site. During this filter change the filter housing shall be cleaned. Overloading of the particulate filter may change the concentration of carbon monoxide.

[d] Dependent on site-specific conditions.

[e] If infringement of an action criterion occurs, corrective actions shall be taken as soon as possible. An evaluation of the influence of the detected infringement on the measurement data produced before the actual correction of the infringement took place shall be given and taken into account during data validation. To ensure that the data capture criterion is met, data will need to be inspected by a trained operator every working day.

9.5 Calibration of the analyser

Calibration shall be performed at least every three months at a recommended concentration of 70 % to 80 % of the certification range, to determine analyser response and drift. Calibration every two weeks will give a better indication of drift and analyser performance.

Carbon monoxide calibration gases traceable to national standards shall be used for the correct calibration of the analyser. Maximum permitted uncertainty in the concentration of gases used for ongoing quality control is ± 5 % with a level of confidence of 95 %. The zero gas shall give no instrument reading higher than the detection limit.

Calibration gases shall be introduced before the filter to check and if necessary to correct for contamination of the filter.

Calibration gases shall be introduced for at least 10 min before any valid readings are taken from the data acquisition system.

Frequency of test:

At least every three months and after repair.

Action criteria:

If zero or span calibration response have drifted beyond the measurement range in use.

Appropriate action:

Adjust or service the analyser

Analysers should not be adjusted in the field unless strictly necessary. All drifts should be taken into account when processing the data. Analysers should only be adjusted if the response to either span or zero gas has drifted beyond the end of the valid measurement range of the analyser.

NOTE This is because some analysers take a very long time to stabilise after adjustment; also some site operatives do not correctly record these adjustments. In both these cases large amounts of data have to be considered invalid.

9.6 Checks

9.6.1 Test gases

For the calibration as well as the zero and span checks of the analyser, several methods are available to generate calibration gases. In Table 3 the various methods are given. For each of the methods the user shall demonstrate that the uncertainty of the calibration gas does not add to the expanded uncertainty budget in such a way that the expanded uncertainty for the method is exceeded.

Gases for span and zero checks shall be verified at least every six months with use of reference gases traceable to national standards. The zero gas shall give no instrument readings higher than the detection limit, and the gas used for daily span checks shall not differ by more than 5 % of the last certified value.

The purity of the gases shall be as specified in Table 4.

Action criteria:

Zero ≥ detection limit;

Span ≥ 5 % of previous verification.

Appropriate action:

Replace the cylinder.

9.6.2 Zero and span checks

Zero and span gas can be supplied by gas cylinder, or generated by an external calibrator unit or internally in the analyser.

Zero and span gas should be introduced into the analyser for a period of 10 min per gas. The timing of this injection shall coincide with a discrete 10-minutes average collection by the data acquisition system. The timing should also be arranged so that zero gas and span gas are injected during consecutive hourly periods. This timing regime ensures that each hour still contains 75 % of valid data and therefore these hours do not reduce the final data capture.

The differences between two zero or two span values obtained from the following equations should be calculated to determine if the action criteria have been exceeded:

$$X_z = |Z_i - Z_0| \tag{20}$$

where

X_z is the difference between the readings of the recent zero check and the most recent calibration;

Z_1 is the reading of the recent zero check;

Z_2 is the reading of the most recent calibration.

$$X_s = \frac{|S_i - S_0|}{S_0} \times 100\% \qquad (21)$$

where

X_s is the difference between the readings of the recent span check and the most recent calibration (%);

S_i is the reading of the recent span check;

S_0 is the reading of the most recent calibration.

Frequency of test:

At least every two weeks. Recommended every 23 h or 25 h.

Action criteria:

Zero ≥ 5 nmol/mol of the initial zero span;

Span ≥ 5 % of initial span value.

Appropriate action:

The site should be visited and the correct function of the analyser determined by calibration.

9.6.3 Lack of fit

Delivering a test gas in a field environment, that meets the specifications of the laboratory test, is difficult. As this parameter has already been tested in the laboratory and included in the uncertainty budget, it is not necessary to test this criterion to such a tight specification as for the laboratory tests. It is acceptable to test the lack of fit to a lesser specification (e.g. ± 6 % measured value). If the analyser fails this specification, then the analyser should be removed from site and further, more rigorous testing should occur.

The lack of fit of the analyser shall be tested using the following concentrations: 0 %, 20 %, 60 % and 95 % of the maximum of the certification range. At each concentration (including zero) at least two individual readings shall be performed. After each change in concentration at least four response times shall be taken into account before the next measurement is performed.

Frequency of test:

After installation and after repair the analyser shall be tested within 1 year; if the result of this test is ≤ 2,0 % of the measured value the next test shall be carried out within 3 years; if the result is > 2,0 % but ≤ 6,0 % the analyser shall be retested within 1 year; if the result is > 6,0 % appropriate action shall be taken;

Action criteria:

> 6,0 % of the measured value.

Appropriate action:

Remove analyser from site for further testing and repair if necessary.

NOTE Lack of fit may be checked either in the laboratory or on site.

9.6.4 Testing the sample manifold

9.6.4.1 Procedure for measuring pressure drop induced by the manifold pump

Equipment required: Handheld manometer.

1) Zero manometer with both ports open to atmospheric pressure;

2) Disconnect analyser sample pipework from sample manifold;

3) Attach –ve port of manometer to sample port on manifold, leave +ve port open to atmospheric pressure;

4) Record measured pressure drop in Pa/mbar;

5) Disconnect manometer and reconnect analyser sample pipework.

The resulting pressure drop should be used to calculate the induced effect on the analyser's response using the following equation:

$$\Delta R_{\mathrm{a}} = b_{\mathrm{gp}} \times \Delta P_{\mathrm{m}} \qquad (22)$$

where

ΔR_a is the change in the analyser's response due to the influence of the pressure drop induced by the manifold pump, expressed as a percentage;

b_{gp} is the sensitivity coefficient of the analyser to sample gas pressure change expressed as a percentage of the measured value, obtained during the laboratory type approval test;

ΔP_m is the measured pressure drop induced by the manifold pump.

Action criteria:

Influence of the pressure drop induced by the manifold sampling pump on the measured concentration < 1 %.

Appropriate action:

Reduce flow through manifold to reduce the induced pressure drop so that the criterion in 6.6 is met.

9.6.4.2 Procedure for testing the sample collection efficiency of the sampling system

The flow rate of the test gas in the sampling system should be such that the residence time is greater than or equal to that found under normal operating conditions. Typical manifold systems (diameter ~30 mm, length 2 m) have a volume of ~2 l and shall have a maximum residence time of 5 s.

Frequency of test:

At least every three years.

Action criteria:

Sample loss > 2 %.

Appropriate action:

Clean/replace/repair manifold as necessary and re-test.

NOTE A possible test procedure for the sample collection efficiency is given below.

Test gas for CO analysers is provided by either dynamic dilution of single or multicomponent cylinders, or from gas cylinders without dilution. Zero-grade air, or better, should be used as a diluent/balance gas for these cylinders.

The results from the tests are in the form of ratio measurement and therefore the correct calibration of the analysers is not necessary, nor is the exact concentration of the test gas. The concentration of the test gas need, however, to be stable.

During testing, the analyser output need to be collected through the data collection system at the monitoring site and the normal site operating procedures followed.

Testing comprises two measurements of the test gas, direct sampling of test gas and the sampling of test gas from the complete sampling system. Data averaged over period of 10 min is recorded for each stage of the test.

An example is given in Annex E of possible manifold testing equipment.

Sample system collection efficiency, E_{ss}, is then calculated as follows:

$$E_{ss} = (1 - (\frac{R_d - R_m}{R_d})) \times 100\ \% \tag{23}$$

where

E_{ss} is the sample system collection efficiency;

R_d is the mean analyser response to the test gas directly sampled by the analyser;

R_m is the mean analyser response to the test gas via the sample manifold.

9.7 Maintenance

9.7.1 Change of particulate filters

It is suggested that the effective filter life is determined at the original site installation stage and then calibrations made before a filter is changed. New filters should be conditioned with ambient air for 30 min before any ambient data is considered valid. This conditioning time should not be included in any calculation of data capture.

Frequency of test:

Determined from initial site installation, strongly dependent on site conditions.

Action criteria:

Response to span gas passing the filter is ≤ 97 %.

Appropriate action:

Calibrate instruments with site standard before changing filter. New filters should be conditioned with ambient air for 30 min before any ambient data is considered valid.

9.7.2 Change of consumables as applicable

The life of all analyser consumables should be determined at the initial installation. Site-specific maintenance periods should be devised for the replacement of such consumables.

9.7.3 Regular maintenance of components of the analyser

The manufacturers recommendations should be followed for the routine maintenance of the analyser.

9.8 Data handling and data reports

The designated body responsible for QA/QC of the monitoring station has the responsibility to produce valid data. This implies that the collected data shall be free from faulty data, zero and span checks or calibrations. When negative values are reported these values shall be included in the calculation of averages. When a value that is larger than the maximum of the certification range is reported, then this value shall be used in the calculation of averages and the said average shall be flagged in such a way that it is clear in the data report that this average might have exceeded the combined uncertainty requirement.

Data collection from the analyser shall be done at a frequency of at least twice per response time period.

Data capture shall be ≥ 75 % of the averaging time.

The response of the analyser to zero or span test gases should be recorded before and after the adjustment of the analyser. Measured drift in the analyser's response that is less than the action criteria in Table 7 shall be corrected for in data processing and not by physical adjustment of the analyser.

The uncertainty of measurement shall be determined and reported every year during operation, e.g. yearly verification or evaluation of data of QA/QC measures. When doing this, as far as possible, the actual values of the performance characteristics obtained during the type approval test shall be used. The specifications of ENV 13005 and EN ISO 14956 shall be used.

10 Expression of results

The readings from the analyser are converted to concentrations using the appropriate conversion factors and the results expressed in milligrams per cubic metre.

The conversion factors at 20 °C and 101,3 kPa are:

1 mg CO/m^3 = $0{,}86 \times 10^{-6}$ (ppm) (V/V) CO= 0,86 µmol/mol CO

1×10^{-6} (ppm) (V/V) CO = 1 µmol/mol CO = 1,16 mg CO/m^3

11 Test reports and documentation

11.1 Type approval tests

The designated body shall prepare a type approval report, which contains at least the following information:

a) complete identification of a each specific type of analyser of a series, including software version;

b) specification of the type approval tests executed e.g. by reference to this document;

c) analyser settings during type approval test;

d) series of observations obtained in the type approval tests;

e) values determined for the performance characteristics evaluated and the applied evaluation methods (e.g. by reference to this document);

f) calculation of the expanded uncertainty of 8-hour mean averages.

The type approval report shall be available to the public.

11.2 Field operation

11.2.1 Suitability evaluation

The user and/or operator of an analyser or monitoring station shall prepare a report of the suitability evaluation of the analyser at the monitoring site. The suitability test report shall contain at least the following information:

a) reference to this document;

b) complete identification of the analyser and monitoring station;

c) results of the suitability evaluation of the analyser at the monitoring site;

d) proof of the compliance of the sampling system to this document;

e) checks on the initial installation.

11.2.2 Documentation

The user and/or operator shall document all maintenance, repairs, calibrations, change of calibration gases, malfunctionings, etc. for each individual analyser, sampling system and monitoring site.

11.2.3 Ambient air quality data reports

Ambient air data reports for carbon monoxide should be prepared according to local, national or EU-regulations (200/69/EC 98/0333 (COD)). The report shall contain at least the following information:

a) reference to this document;

b) percentage of data capture;

c) air quality data presented in the required format;

d) a statement on measurement uncertainty of the data reported.

Annex A
(normative)

Calculation of lack of fit

A.1 Establishment of the regression line

A linear regression function in the form of $Y_i = A + B \times X_i$ is made though calculation of the function $Y_i = a + B\,(X_i - X_z)$.

For the regression all measuring points (including zero) are taken into account. The total number of measuring points (n) is equal to the number of concentration levels (at least six including zero) times the number of repetitions (at least five) at a particular concentration level.

The coefficient a is obtained from:

$$a = \sum Y_i / n \tag{A.1}$$

where

a is the average value of the Y-values;

Y_i is the individual Y-value;

n is the number of calibration points.

The coefficient B is obtained by:

$$B = \left(\sum Y_i \left(X_i - X_z\right)\right) / \sum \left(X_i - X_z\right)^2 \tag{A.2}$$

where

X_z is the average of the X-values ($= \Sigma X_i/n$);

X_i is the individual X-value.

The function $Y_i = a + B\,(X_i - X_z)$ is converted to $Y_i = A + B \times X_i$ through the calculation of A:

$$A = a - B \times X_z \tag{A.3}$$

A.2 Calculation of the residuals of the averages

The residuals of the averages of each calibration point (including the zero point) are calculated as follows:

The average of each calibration point (including the zero point) at one and the same concentration c is calculated according to:

$$(Y_a)_c = \Sigma\,(Y_i)_c / m \tag{A.4}$$

where

$(Y_a)_c$ is the average Y-value at concentration level c;

$(Y_i)_c$ is the individual Y-value at concentration level c;

m is the number of repetitions at one and the same concentration level $c = (\Sigma\, Y_i/m)_c$.

The residual of each average (d_c) at each concentration level c is calculated according to:

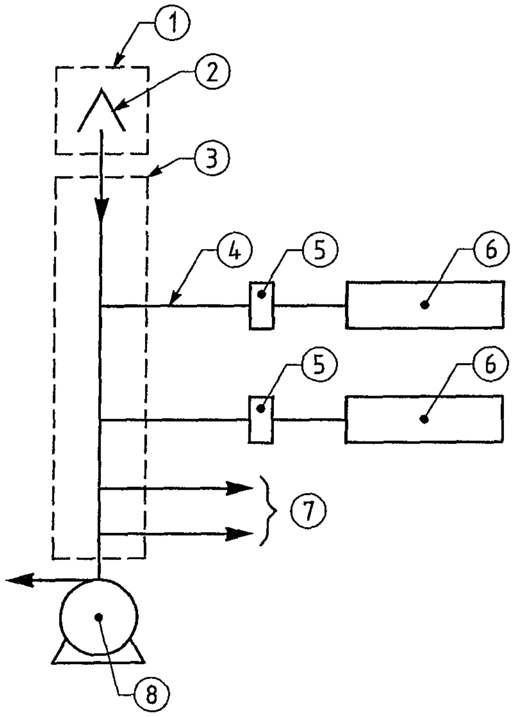
$$d_c = (Y_a)_c - (A + B \times c) \tag{A.5}$$

Each residual to a value relative to its own concentration level c is expressed as:

$$(d_r)_c = \frac{d_c}{c} \times 100\ \% \tag{A.6}$$

Annex B
(informative)

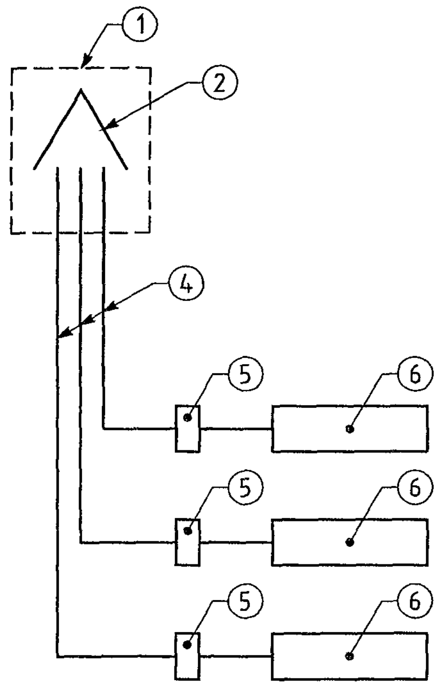
Sampling equipment



Key

1 Sample inlet
2 Rain shield
3 Manifold
4 Sampling line
5 Filter
6 Analyser
7 Connection for other analysers
8 Sampling pump for the manifold

Figure B.1 — Sampling layout with a main sampling manifold



Key

1 Sample inlet
2 Rain shield
4 Sampling line
5 Filter
6 Analyser

Figure B.2 — Sampling layout with individual lines

Annex C
(informative)

Sampling on micro scale

The following guidelines should be met as far as practicable:

— the flow around the inlet sampling probe should be unrestricted without any obstructions affecting the airflow in the vicinity of the sampler (normally some metres away from buildings, balconies, trees and other obstacles and at least 0,5 m away from the nearest building in the case of sampling points representing air quality at the building line);

— in general, the inlet sampling points should be between 1,5 m (the breathing zone) and 4 m above the ground. Higher positions (up to 8 m) may be necessary in some circumstances. Higher siting may also be appropriate if the station is representative of a large area;

— the inlet probe should not be positioned in the immediate vicinity of sources in order to avoid the direct intake of emissions unmixed with ambient air;

— the sampler's exhaust outlet should be positioned so that recirculation of exhaust air to the sampler inlet is avoided;

— location of traffic orientated samplers:

  — for all pollutants, such sampling points should be at least 25 m from the edge of major junctions and at least 4 m from the centre of the nearest traffic lane;

  — for nitrogen dioxide, inlets should be no more than 5 m from the kerbside;

  — for particulate matter and lead, inlets should be sited so as to be representative of air quality near to the building line.

The following factors may also be taken into account:

— interfering sources;

— security;

— access;

— availability of electrical power and telephone communications;

— visibility of the site in relation to its surroundings;

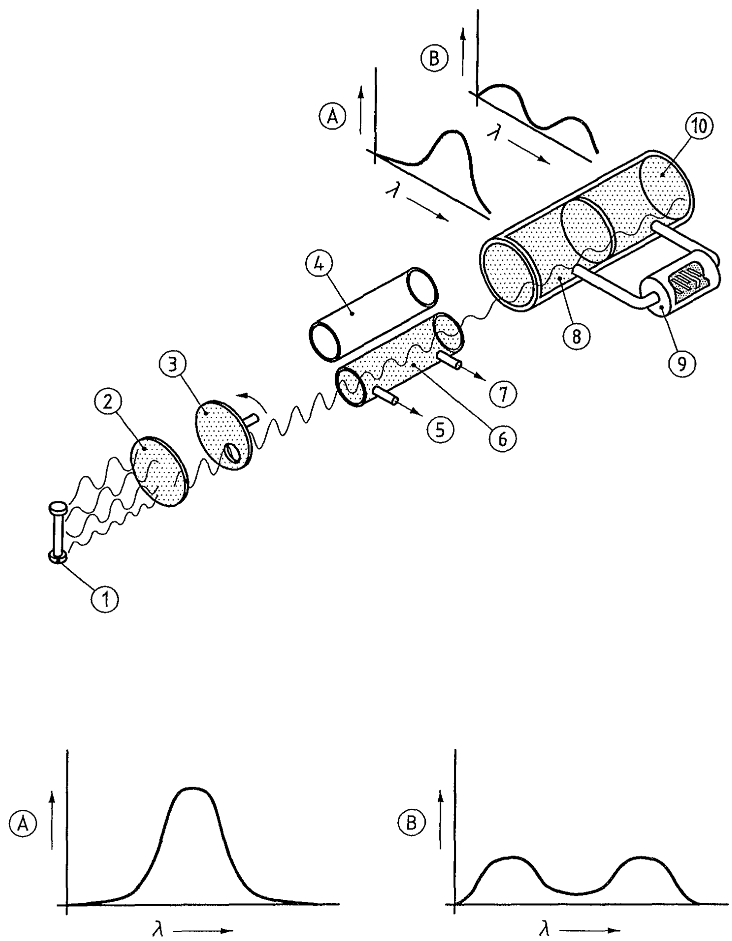
— safety of public and operators;

— the desirability of co-locating sampling points for different pollutants;

— planning requirements.

Annex D
(informative)

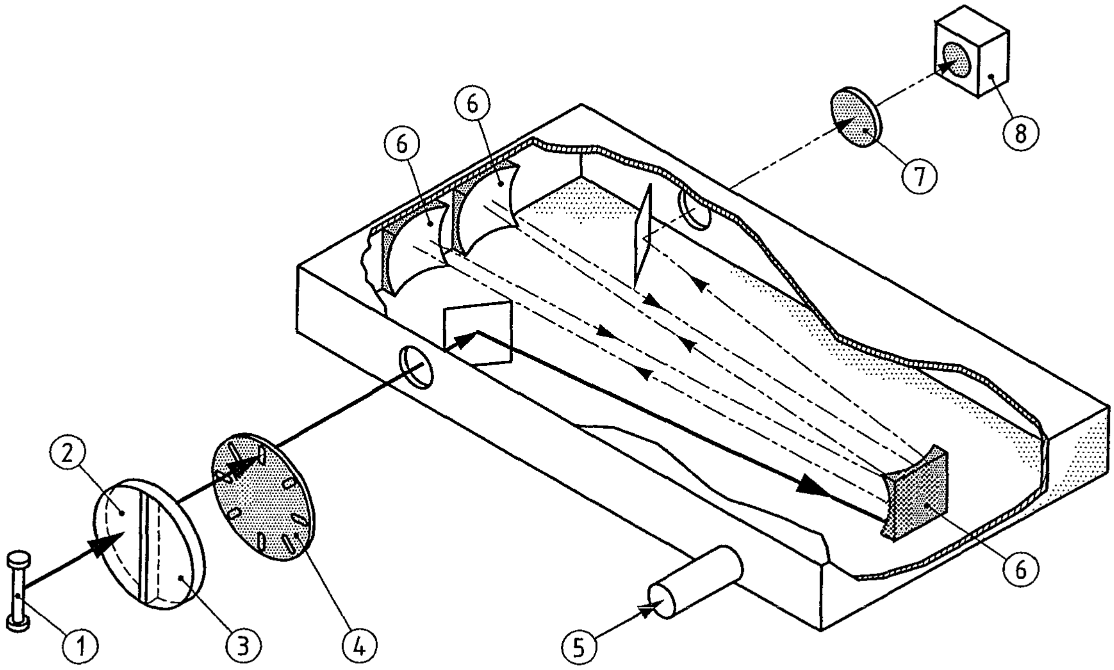
Schematics of non-dispersive infrared spectrometer



Key

1 IR lamp
2 Light filter
3 Chopper
4 Reference cell
5 Gas out
6 Sample cell
7 Gas out
8 First chamber
9 Microflow sensor
10 Second chamber
A Light absorption in first chamber
B Light absorption in second chamber
λ Wavelength

Figure D.1 – Example of a dual cell analyser



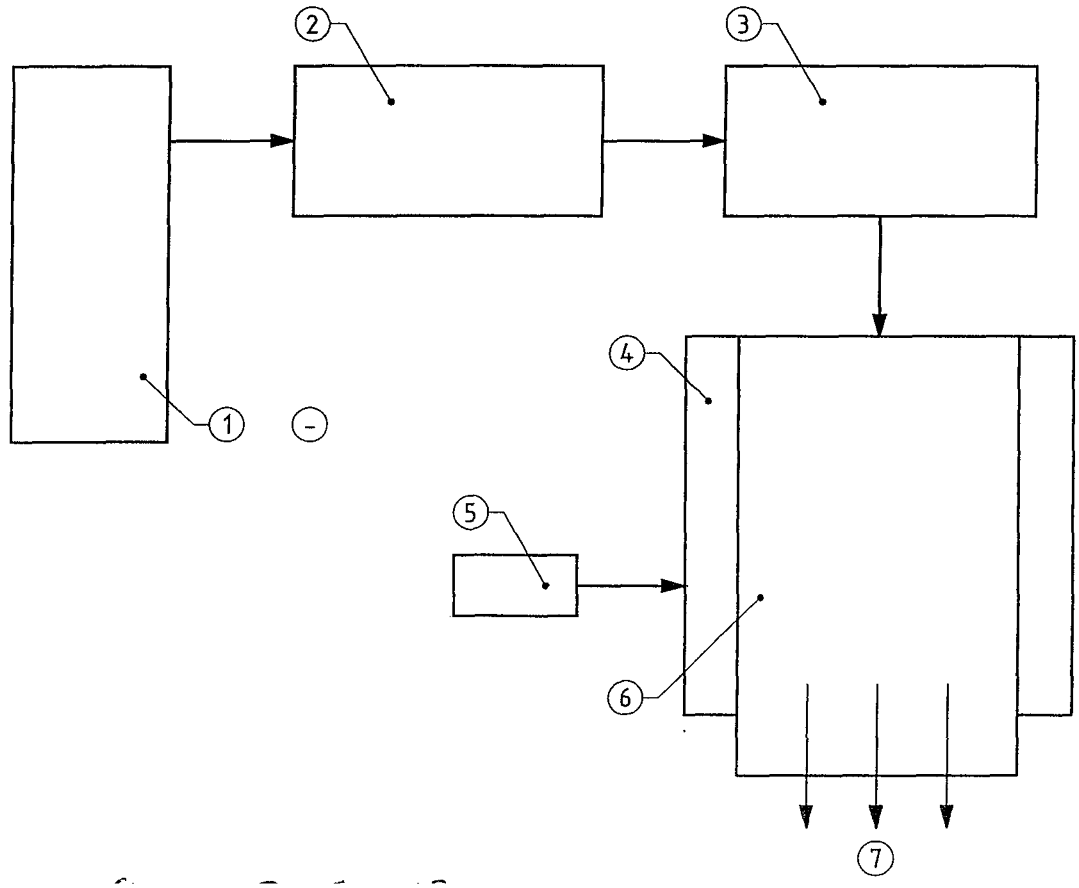
Key

1 IR source
2 Neutral filter (N_2)
3 Gas filter cell (CO)
4 Modulator
5 Sample gas
6 Mirror
7 Filter
8 Detector

Figure D.2 – Example of a gas filter correlation analyser

Annex E
(informative)

Manifold testing equipment



Key

1 Gas cylinder, 200 nmol/mol NO_2, 150 nmol/mol SO_2, 5 µmol/mol CO, balanced air
2 50 l/min mass flow controller
3 Ozone generator (producing 100 nmol/mol at 50 l/min)
4 Tedlar bag, secured around manifold
5 Pressure relief valve
6 Air sampling manifold
7 Analysers

Figure E.1 – Schematic Manifold testing equipment

Annex F
(normative)

Type approval

F.1 Type approval and uncertainty calculation

F.1.1 Type approval

The type approval of the analyser consists of the following steps (see Clause 8):

1) the value of each individual performance characteristic tested in the laboratory shall fulfil the criterion stated in Table 1 (see 8.2);

2) the expanded uncertainty calculated from the standard uncertainties due to the values of the specific performance characteristics obtained in the laboratory tests fulfils the criterion as stated in the Council Directive 2000/69/EC. This criterion is the maximum uncertainty of hourly values of continuous measurements at the 8-hour mean limit value. The relevant specific performance characteristics and the calculation procedure are given in this annex;

3) the value of each of the individual performance characteristics tested in the field shall fulfil the criterion stated in Table 1 (see 8.2);

4) the expanded uncertainty calculated from the standard uncertainties due to the values of the specific performance characteristics obtained in the laboratory and field tests fulfils the criterion as stated in the Council Directive 2000/69/EC. This criterion is the maximum uncertainty of hourly values of continuous measurements at the 8-hour mean limit value. The relevant specific performance characteristics and the calculation procedure are given in this annex.

The instrument is type approved when all 4 requirements are met.

F.1.2 Uncertainty calculation

The model equation for the measurement of CO is based on the assumption that the measured value c consists of signal contribution from the concentration of CO, c', and a sum of signal contributions c_K (in units of the measured value) caused by the influence of the considered performance characteristics K of the measuring system:

$$c = c' + \sum_K c_K \tag{F.1}$$

From this model equation the following variance equation is derived which describes the uncertainty of the measured value, $u(c)$, under the conditions of the type approval test:

$$u^2(c) = \sum_K u^2(c_K) \tag{F.2}$$

Equation (F.3) is used for estimating the combined standard uncertainty $u(c)$ in type approval step 2. For type approval step 4, which includes the results of the field tests, Equation (F.18) is used for estimating the combined standard uncertainty $u(c)$.

F.2 Type approval requirement 1

Table F.1 gives the performance characteristics that shall be considered in demonstrating compliance with requirement 1 in the type approval procedure and an example of laboratory results.

Table F.1 — Relevant performance characteristics and criteria

No.	Performance characteristic	Symbol	Clause	Performance criterion for CO		Results lab. Test [b]	[c]
1	Repeatability standard deviation at zero	$s_{r,z}$	8.4.5	≤ 1,0	µmol/mol	0,40	✓
2	Repeatability standard deviation at concentration c_t (at a level of the 8-hour mean limit value)	$s_{r,ct}$	8.4.5	≤ 3,0	µmol/mol	2,0	✓
3	Lack of fit (residual from the linear regression function		8.4.6				
3a	Largest residual from the linear regression function at concentrations higher than zero	X_l		≤ 4,0	% of the measured value	2,5	✓
3b	Residual at zero	$X_{l,z}$		≤ 0,20	µmol/mol	0,12	✓
4	Sensitivity coefficient of sample gas pressure	b_{sp}	8.4.7	≤ 0,70	µmol/mol /kPa	0,15	✓
5	Sensitivity coefficient of sample gas temperature	b_{gt}	8.4.8	≤ 0,30	µmol/mol/K	0,10	✓
6	Sensitivity coefficient of surrounding temperature	b_{st}	8.4.9	≤ 0,30	µmol/mol/K	0,10	✓
7	Sensitivity coefficient of electrical voltage	b_V	8.4.10	≤ 0,30	µmol/mol/V	0,05	✓
8	Interferents at zero and concentration c_t (at a level of the 8-hour mean limit value)		8.4.11				
8a	H_2O with concentration 19 mmol/mol [d]	$X_{H2O,z}$		≤ 1,0	µmol/mol	0,22	✓
		$X_{H2O,ct}$		≤ 1,0	µmol/mol	0,30	✓
8b	CO_2 with concentration 500 µmol/mol	$X_{CO2,z}$		≤ 0,50	µmol/mol	0,10	✓
		$X_{CO2,ct}$		≤ 0,50	µmol/mol	0,15	✓
8c	NO with concentration 1 µmol/mol	$X_{NO,z}$		≤ 0,50	µmol/mol	0,080	✓
		$X_{NO,ct}$		≤ 0,50	µmol/mol	0,10	✓
8d	N_2O with concentration 50 nmol/mol	$X_{N2O,z}$		≤ 0,50	µmol/mol	0,060	✓
		$X_{N2O,ct}$		≤ 0,50	µmol/mol	0,11	✓
9	Averaging effect	X_{av}	8.4.12	≤ 7,0	% of the measured value	3,0	✓

No.	Performance characteristic	Symbol	Clause	Performance criterion for CO		Results lab. Test [b]	[c]
13	Short term drift at zero	$D_{s,z}$	8.4.4	≤ 0,10	µmol/mol over 12 h	0,75	✓
14	Short term drift at span level [a]	$D_{s,s}$	8.4.4	≤ 0,60	µmol/mol over 12 h	3,5	✓
15	Response time (rise)	t_r	8.4.3	≤ 180	s	36	✓
16	Response time (fall)	t_f	8.4.3	≤ 180	s	33	✓
17	Difference rise time and fall time	t_d	8.4.3	≤ 10	% relative difference	8,3	✓
18	Difference sample/calibration port [e]	D_{sc}	8.4.13	≤ 1,0	%	0,50	✓

[a] span level is 70 % to 80 % of the certification range

[b] to demonstrate compliance the absolute value of the performance characteristic shall be taken

[c] ✓ : requirement is met

[d] a H_2O-concentration of 19 mmol/mol is about 80 % RH at 20 °C and 101,3 kPa

[e] if relevant

CONCLUSION: All values of the performance characteristics obtained in the laboratory tests are complying with the requirements.

Requirement 1 is fulfilled.

F.3 Type approval requirement 2

The values of the following uncertainties shall be included in the calculation of the expanded uncertainty after the laboratory tests.

Table F.2 — Standard uncertainties to be incorporated in the calculation of the expanded uncertainty after the laboratory tests

No.	Standard uncertainty due to	Symbol	Equation, F.
1	Repeatability standard deviation at zero	$u_{r,z}$	6, 7
2	Repeatability standard deviation at the 8-hour mean limit value	$u_{r,lv}$	8, 9, 10
3	Lack of fit at the 8-hour mean limit value	$u_{l,lv}$	11
4	Variation in sample gas pressure at the 8-hour mean limit value	u_{gp}	12, 13
5	Variation in sample gas temperature at the 8-hour mean limit value	u_{gt}	14, 15
6	Variation in surrounding temperature at the 8-hour mean limit value	u_{st}	16, 17
7	Variation in electrical voltage at the 8-hour mean limit value	u_V	18, 19
8	Interferences at the 8-hour mean limit value		
8a	H_2O with concentration 19 mmol/mol	u_{H2O}	20, 21, 22, 23
8b	CO_2 with concentration 500 µmol/mol	u_{CO2}	24, 25
8c	NO with concentration 1 µmol/mol	u_{NO}	24, 25
8d	N_2O with concentration 50 nmol/mol	u_{N2O}	24, 25
9	Averaging error	u_{av}	30
18	Difference sample/calibration port	uu_{Dsc}	32

Next to the uncertainties due to these performance characteristics the uncertainty in the concentration of calibration gas and in the calibration itself are incorporated in the uncertainty calculation.

Table F.3 — Standard uncertainty of the calibration gas to be incorporated in the calculation of the expanded uncertainty after the laboratory tests

No.	Standard uncertainty due to	Symbol	Equation, F.
21	Uncertainty in calibration gas	u_{cg}	31

The calculation of standard uncertainties is based on the procedures laid down in EN ISO 14956.The uncertainty calculation shall be carried out with the values of the performance characteristics at the concentration of the 8-hour mean limit value (if relevant).

The absolute **expanded uncertainty**, U_c, shall be calculated according to:

$$U_c = k \times u_c \tag{F.3}$$

with $k = 2$

where

U_c is the expanded uncertainty (µmol/mol);

k is the coverage factor of approximately 95 %;

u_c is the combined standard uncertainty (µmol/mol);

The relative **expanded uncertainty**, $U_{c,rel}$, shall be calculated according to:

$$U_{c,rel} = (U_c / hlv) \times 100\,\% \tag{F.4}$$

where

$U_{c,rel}$ is the relative expanded uncertainty (%);

U_c is the expanded uncertainty (µmol/mol);

hlv is the 8-hour mean limit value (µmol/mol);

Requirement 2 is fulfilled when: $U_{c,rel} \leq U_{req,rel}$.

The **combined standard uncertainty**, u_c, shall be calculated according to:

$$u_c = \sqrt{u_{r,z}^2 + u_{r,lv}^2 + u_{l,lv}^2 + u_{gp}^2 + u_{gt}^2 + u_{st}^2 + u_v^2 + u_{H_2O}^2 + (u_{int,pos}^2 \text{ or } u_{int,neg}^2) + u_{av}^2 + u_{Dsc}^2 + u_{cg}^2} \tag{F.5}$$

where

u_c is the combined standard uncertainty (µmol/mol);

$u_{r,z}$ is the standard uncertainty for repeatability at zero (µmol/mol);

$u_{r,lv}$ is the standard uncertainty for repeatability at level of the 8-hour mean limit value (µmol/mol);

$u_{l,lv}$ is the standard uncertainty for lack of fit at level of the 8-hour mean limit value (µmol/mol);

u_{gp} is the standard uncertainty for sample gas pressure variation (µmol/mol);

u_{gt} is the standard uncertainty for sample gas temperature variation (µmol/mol);

u_{st} is the standard uncertainty for surrounding temperature variation (µmol/mol);

u_V is the standard uncertainty for electrical voltage variation (µmol/mol);

u_{H2O} is the standard uncertainty for the presence of water vapour (µmol/mol);

$u_{int,pos}$ is the standard uncertainty for interferents (except water vapour) with positive impact (µmol/mol);

$u_{int,neg}$ is the standard uncertainty for interferents (except water vapour) with negative impact (µmol/mol);

u_{av} is the standard uncertainty for averaging (µmol/mol);

u_{Dsc} is the standard uncertainty for difference sample/calibration port (µmol/mol);

u_{cg} is the standard uncertainty for calibration gas (µmol/mol).

For the calculation of the standard uncertainties the following equations shall be used.

The standard uncertainty for **repeatability at zero**, $u_{r,z}$, is calculated according to:

$$u_{r,z} = s_{r,z} / \sqrt{m} \tag{F.6}$$

with

$$m = t / ((t_r + t_f) / 2) \tag{F.7}$$

where

$u_{r,z}$ is the standard uncertainty for repeatability at zero (µmol/mol);

t is 3 600 s;

t_r is the response time (rise) (s);

t_f is the response time (fall) (s);

$s_{r,z}$ is the repeatability standard deviation at zero (µmol/mol);

m is the number of individual measurements in 3 600 s, taking response (rise) and response time (fall) for half that time period respectively.

The standard uncertainty for **repeatability at the 8-hour mean limit value**, $u_{r,lv}$, is calculated according to:

$$u_{r,lv} = s_{r,lv} / \sqrt{m} \tag{F.8}$$

with

$$m = t / ((t_r + t_f) / 2) \tag{F.9}$$

and

$$s_{r,lv} = (hlv / c_t) \times s_{r,ct} \tag{F.10}$$

where

$u_{r,lv}$ is the standard uncertainty for repeatability at the 8-hour mean limit value (µmol/mol);

$s_{r,lv}$ is the repeatability standard deviation at the 8-hour mean limit value (µmol/mol);

m is the number of individual measurements in 3 600 s, taking response (rise) and response time (fall) for half that time period respectively;

t is 3 600 s;

t_r is the response time (rise) (s);

t_f is the response time (fall) (s);

hlv is the 8-hour mean limit value (µmol/mol);

c_t is the test gas concentration (at the level of the 8-hour mean limit value) (µmol/mol);

$s_{r,ct}$ is the repeatability standard deviation at the test gas concentration (µmol/mol).

The standard uncertainty due to **lack of fit at the 8-hour mean limit value**, $u_{l,lv}$, is calculated according to:

$$u_{l,lv} = ((X_{l,lv} / 100) \times hlv) / \sqrt{3} \quad \text{(F.11)}$$

where

$u_{l,lv}$ is the standard uncertainty due to lack of fit at the 8-hour mean limit value (µmol/mol);

$X_{l,lv}$ is the calculated residual from a linear regression function at the 8-hour mean limit value (%) (calculated according to Annex A);

hlv is the 8-hour mean limit value (µmol/mol).

The standard uncertainty due to variation of **sample gas pressure at the 8-hour mean limit value**, u_{sp}, is calculated according to:

$$u_{gp} = ((hlv/c_t) \times b_{gp}) \times (\Delta_{gp} / \sqrt{3}) \quad \text{(F.12)}$$

with

$$\Delta_{gp} = P_1 - P_2 \quad \text{(F.13)}$$

where

u_{gp} is the standard uncertainty due to the influence of pressure (µmol/mol);

b_{gp} is the sensitivity coefficient of sample gas pressure variation (%/kPa);

hlv is the 8-hour mean limit value (µmol/mol);

c_t is the test gas concentration (70 % to 80 % of the certification range) (µmol/mol);

Δ_{gp} is the pressure range, used in the laboratory test (kPa);

P_1 is the sample gas pressure P_1 (kPa);

P_2 is the sample gas pressure P_2 (kPa).

The standard uncertainty due to variation of **sample gas temperature at the 8-hour mean limit value**, u_{gt}, is calculated according to:

$$u_{gt} = ((hlv / c_t) \times b_{gt}) \times (\Delta_{gt} / \sqrt{3}) \quad \text{(F.14)}$$

with

$$\Delta_{gt} = T_{gt,1} - T_{gt,2} \quad \text{(F.15)}$$

where

u_{gt} is the standard uncertainty due to the variation of sample gas temperature (µmol/mol);

hlv is the 8-hour mean limit value (µmol/mol);

c_t is the test gas concentration (70 % to 80 % of the certification range) (µmol/mol);

b_{gt} is the sensitivity coefficient of sample gas temperature variation (µmol/mol);

Δ_{gt} is the sample gas temperature range, used in the laboratory test (K);

$T_{gt,1}$ is the sample gas temperature T_1 (in K);

$T_{gt,2}$ is the sample gas temperature T_2 (in K).

The standard uncertainty due to variation of **surrounding temperature at the 8-hour mean limit value**, u_{st}, is calculated according to:

$$u_{st} = ((hlv / c_t) \times b_{st}) \times (\Delta_{st} / \sqrt{12}) \quad \text{(F.16)}$$

with

$$\Delta_{st} = T_{st,1} - T_{st,2} \quad \text{(F.17)}$$

where

u_{st} is the standard uncertainty due to the variation of the surrounding temperature (µmol/mol);

hlv is the 8-hour mean limit value (µmol/mol);

c_t is the test gas concentration (70 % to 80 % of the certification range) (µmol/mol);

b_{st} is the sensitivity coefficient of the surrounding temperature variation (µmol/mol/K);

Δ_{st} is the surrounding temperature range, used in the laboratory test (K);

$T_{st,1}$ is the surrounding temperature T_1 (K);

$T_{st,2}$ is the surrounding temperature T_2 (K).

The standard uncertainty due to variation of **electrical voltage at the 8-hour mean limit value**, u_V, is calculated according to:

$$u_v = ((hlv / c_t) \times b_v) \times (\Delta_v / \sqrt{12}) \quad \text{(F.18)}$$

with

$$\Delta_v = V_1 - V_2 \quad \text{(F.19)}$$

where

u_V is the standard uncertainty due to the variation of electrical voltage (µmol/mol);

hlv is the 8-hour mean limit value (µmol/mol);

c_t is the test gas concentration (70 % to 80 % of the certification range) (µmol/mol);

b_V is the sensitivity coefficient of electrical voltage variation (µmol/mol/V);

Δ_V is the electrical voltage range, used in the laboratory test (V);

V_1 is the electrical voltage V_1 (V);

V_2 is the electrical voltage V_2 (V).

The influence quantity of water vapour is established at a water concentration of 19 mmol/mol. The uncertainty, however, is to be established at a water concentration of 21 mmol/mol. The standard uncertainty due to **interference by the presence of water vapour at the 8-hour mean limit value**, u_{H2O}, is therefore calculated according to:

$$X_{\mathrm{H_2O,z,max}} = (18/16) \times X_{\mathrm{H_2O,z}} \quad \text{(F.20)}$$

$$X_{\mathrm{H_2O,c_t,max}} = (18/16) \times X_{\mathrm{H_2O,c_t}} \quad \text{(F.21)}$$

$$X_{\mathrm{H_2O,max}} = ((X_{\mathrm{H_2O,c_t,max}} - X_{\mathrm{H_2O,z,max}}) / c_\mathrm{t}) \times hlv + X_{\mathrm{H_2O,z,max}} \quad \text{(F.22)}$$

$$u_{\mathrm{H_2O}} = \left| X_{\mathrm{H_2O}} / c_{\mathrm{H_2O,max}} \right| \times \sqrt{(c_{\mathrm{H_2O,max}}{}^2 + c_{\mathrm{H_2O,max}} \times c_{\mathrm{H_2O,min}} + c_{\mathrm{H_2O,min}}{}^2)/3} \quad \text{(F.23)}$$

where

$X_{\mathrm{H_2O,z,max}}$ is the influence quantity of an H_2O concentration of 21 mmol/mol at zero concentration of the measurand (µmol/mol);

$X_{\mathrm{H_2O,z}}$ is the influence quantity of an H_2O concentration of 19 mmol/mol at zero concentration of the measurand (µmol/mol);

$X_{\mathrm{H_2O,c_t,max}}$ is the influence quantity of an H_2O concentration of 21 mmol/mol at the test concentration c_t of the measurand (µmol/mol);

$X_{\mathrm{H_2O,c_t}}$ is the influence quantity of an H_2O concentration of 19 mmol/mol at the test concentration c_t of the measurand (µmol/mol);

$X_{\mathrm{H_2O}}$ is the influence quantity of an H_2O concentration of 21 mmol/mol at the 8-hour mean limit value (µmol/mol);

c_t is the test gas concentration of the measurand (µmol/mol);

hlv is the 8-hour mean limit value (µmol/mol);

$u_{\mathrm{H_2O}}$ is the standard uncertainty due to interference by the presence of water vapour (µmol/mol);

$c_{\mathrm{H_2O,max}}$ is the maximum concentration of water vapour (mmol/mol) (= 218 mmol/mol);

$c_{\mathrm{H_2O,min}}$ is the minimum concentration of water vapour (mmol/mol) (= 6 mmol/mol).

The standard uncertainty due to each **interfering compound at the 8-hour mean limit value** (other than water vapour), u_{int}, is calculated according to:

$$X_{\mathrm{int}} = ((X_{\mathrm{int,c_t}} - X_{\mathrm{int,z}}) / c_\mathrm{t}) \times hlv + X_{\mathrm{int,z}} \quad \text{(F.24)}$$

$$u_{\text{int}} = \left| X_{\text{int}} / c_{\text{int,max}} \right| \times \sqrt{(c_{\text{int,max}}^{2} + c_{\text{int,max}} \times c_{\text{int,min}} + c_{\text{int,min}}^{2})/3} \tag{F.25}$$

where

$X_{\text{int,c}_t}$ is the influence quantity of the maximum concentration (nmol/mol resp. µmol/mol) of the relevant interfering compound at the test concentration c_t of the measurand (µmol/mol)

$X_{\text{int,z}}$ is the influence quantity of the maximum concentration (nmol/mol resp. µmol/mol) of the relevant interfering compound at zero concentration of the measurand (µmol/mol);

X_{int} is the influence quantity of the relevant interfering compound (µmol/mol);

c_t is the test concentration of the measurand at the level of the 8-hour mean limit value (µmol/mol);

hlv is the 8-hour mean limit value (µmol/mol);

u_{int} is the standard uncertainty due to interference by the presence of a chemical compound (µmol/mol);

$c_{\text{int,max}}$ is the maximum concentration of interfering compound (nmol/mol resp. µmol/mol);

$c_{\text{int,min}}$ is the minimum concentration of interfering compound (nmol/mol resp. µmol/mol).

According to EN ISO 14956 the summed uncertainties due to the interferents with positive impact and the summed uncertainties of the interferents with negative impact shall be calculated.

$$S_{u_{\text{int,pos}}} = u_{\text{int,1,pos}} + u_{\text{int,2,pos}} + \ldots\ldots + u_{\text{int,n,pos}} \tag{F.26}$$

$$S_{u_{\text{int,neg}}} = u_{\text{int,1,neg}} + u_{\text{int,2,neg}} + \ldots\ldots + u_{\text{int,n,neg}} \tag{F.27}$$

Take the highest sum as the representative value for all interferents.

$$u_{\text{int,pos}} = \sqrt{(u_{\text{int,1,pos}} + u_{\text{int,2,pos}} + \ldots\ldots + u_{\text{int,n,pos}})^{2}} \tag{F.28}$$

$$u_{\text{int,neg}} = \sqrt{(u_{\text{int,1,neg}} + u_{\text{int,2,neg}} + \ldots\ldots + u_{\text{int,n,neg}})^{2}} \tag{F.29}$$

where

$u_{\text{int,pos}}$ is the sum of uncertainties due to interferents with positive impact (µmol/mol);

$u_{\text{int,1,pos}}$ is the uncertainty due to the 1th interferent with positive impact (µmol/mol);

$u_{\text{int,n,pos}}$ is the uncertainty due to the *n*th interferent with positive impact (µmol/mol);

$u_{\text{int,neg}}$ is the sum of uncertainties due to interferents with negative impact (µmol/mol);

$u_{\text{int,1,neg}}$ is the uncertainty due to the 1th interferent with negative impact (µmol/mol);

$u_{\text{int,n,neg}}$ is the uncertainty due to the *n*th interferent with negative impact (µmol/mol).

The standard uncertainty due to the **averaging error**, u_{av}, is calculated according to:

$$u_{\text{av}} = ((X_{\text{av}} / 100) \times hlv) / \sqrt{3} \tag{F.30}$$

where

u_V is the standard uncertainty due to the averaging error (µmol/mol);

X_{av} is the averaging error (% of the measured value);

hlv is the 8-hour mean limit value (µmol/mol).

The standard uncertainty due to the **calibration gas**, u_{cg}, is calculated according to:

$$u_{cg} = ((X_{cg} / 100) \times hlv) / 2 \tag{F.31}$$

where

u_{cg} is the standard uncertainty due to the calibration gas (µmol/mol);

X_{cg} is the error in the calibration gas (%);

hlv is the 8-hour mean limit value (µmol/mol).

The standard uncertainty due to the **difference sample/calibration port**, u_{Dsc}, is calculated according to:

$$u_{D_{sc}} = ((D_{sc} / 100) \times hlv) / \sqrt{3} \tag{F.32}$$

where

$u_{D_{sc}}$ is the standard uncertainty due to the difference sample/calibration port (µmol/mol);

D_{sc} is the difference sample/calibration port (%);

hlv is the 8-hour mean limit value (µmol/mol).

F.4 Example Result of type approval for CO with the results of the laboratory tests

F.4.1 Values of test parameters to be used in the uncertainty calculations

Table F.4 — Physical influences

	test range	lower value	upper value
Sample pressure variation	30 kPa	80 kPa	110 kPa
Sample gas temperature variation	30 K	0 °C	30 °C
Surrounding air temperature variation [a]	30 K	0 °C	30 °C
Voltage variation [a]	40 V	210 V	250 V
[a] specified by the manufacturer			

Table F.5 — Chemical influences/interferents

Interferents		lower value, $c_{int,min}$	upper value, $c_{int,max}$
H_2O	mmol/mol	6	21
CO_2	µmol/mol	0	500
NO	µmol/mol	0	1
N_2O	nmol/mol	0	50

Table F.6 — Test concentrations of the measurand

c_t at a level of the 8-hour mean limit value	9,0 µmol/mol
c_t at span level (70 % to 80 % of the certification range)	65 µmol/mol

Table F.7 — Default values

Uncertainty in calibration gas concentration, expressed at a level of confidence of 95 %	± 3 %

F.4.2 Uncertainty due to interferents, except water vapour (see EN ISO 14956:2002, 8.5.5)

First, calculate the summed uncertainties due to the interferents with positive impact and the summed uncertainties due to the interferents with negative impact. Take the highest sum as the representative value for all interferents.

Table F.8 – Uncertainties due to interferents in the laboratory test

No.	Standard uncertainty due to	Result	Equation F.	uncert. u(p) (µmol/mol)	uncert. $u(p)^2$ $(\mu mol/mol)^2$
8	Interferents				
8b	CO_2 with concentration 500 µmol/mol at zero	+0,10 µmol/mol CO			
	CO_2 with concentration 500 µmol/mol at c_t	+0,15 µmol/mol CO			
	CO_2 with concentration 500 µmol/mol at the 8-hour mean limit value	+0,15 µmol/mol CO	24, 25	0,085	0,007 3
8c	NO with concentration 1 µmol/mol at zero	0,080 µmol/mol CO			
	NO with concentration 1 µmol/mol at c_t	0,10 µmol/mol CO			
	NO with concentration 1 µmol/mol at the 8-hour mean limit value	0,10 µmol/mol CO	24, 25	0,057	0,003 3
8d	N_2O with concentration 50 nmol/mol at zero	0,060 µmol/mol CO			
	N_2O with concentration 50 nmol/mol at c_t	0,11 µmol/mol CO			
	N_2O with concentration 50 nmol/mol at the 8-hour mean limit value	0,11 µmol/mol CO	24, 25	0,062	0,003 9
8b – 8d	summed uncertainty with positive impact		26, 27	0,20	
8b – 8d	summed uncertainty with negative impact		26, 27	0,00	

CONCLUSION: Take the summed uncertainty with positive impact for further calculation.

F.4.3 Calculation of expanded uncertainty from the results of laboratory tests for CO

The following test values have been used for calculating the uncertainties.

Table F.9 — Chemical influences/interferents

		lower value, $c_{H2O,min}$	upper value, $c_{H2O,max}$
H_2O	mmol/mol	6	21

The tests are carried out with the following test concentration of the measurand.

Table F.10 — Test concentration of the measurand

c_t (CO)	µmol/mol	9,0

The test values for sample pressure variation, sample gas temperature variation, surrounding air temperature variation and voltage variation shall be taken. In the calculation in this annex the following values are used in the calculation of the standard uncertainties:

Table F.11 — Values in the calculation of the standard uncertainties

	test range	lower value	upper value
Sample pressure variation	30 kPa	80 kPa	110 kPa
Sample gas temperature variation	30 K	0 °C	30 °C
Surrounding air temperature variation [a]	30 K	0 °C	30 °C
Voltage variation [a]	40 V	210 V	250 V
[a] specification of the manufacturer			

Table F.12 — Default values

Uncertainty in calibration gas concentration, expressed at a level of confidence of 95 %	± 3 %

Table F.13 — Uncertainty from the results of laboratory tests

No.	Standard uncertainty due to	Result or default value		Equation F.	uncert. u(p) (µmol/mol)	uncert. $u(p)^2$ $(µmol/mol)^2$
1	Repeatability standard deviation at zero	0,40	µmol/mol	6, 7	0,04	0,00
2	Repeatability standard deviation at level of the 8-hour mean limit value	2,0	µmol/mol	8, 9, 10	0,19	0,04
3	Lack of fit at the 8-hour mean limit value	2,5	% of the tested value	11	0,12	0,02
4	Sensitivity coefficient of sample gas pressure	0,12	µmol/mol/kPa	12, 13	0,27	0,08
5	Sensitivity coefficient of sample gas temperature	0,10	µmol/mol/K	14, 15	0,23	0,05
6	Sensitivity coefficient of surrounding temperature	0,10	µmol/mol/K	16, 17	0,23	0,05

No.	Standard uncertainty due to	Result or default value		Equation F.	uncert. u(p) (µmol/mol)	uncert. $u(p)^2$ $(\mu mol/mol)^2$
7	Sensitivity coefficient of electrical voltage	0,05	µmol/mol/V	18, 19	0,15	0,02
8	Interferents at level of the 8-hour mean limit value					
	8a H_2O with concentration 21mmol/mol	0,33 [a]	µmol/mol	20, 21, 22, 23	0,23	0,05
	8b - 8d summed uncertainty with positive or negative impact (whichever is the greatest)			24, 25, 26, 27, 28, 29	0,20	0,04
9	Averaging effect	3,0	% of the tested value	30	0,15	0,02
18	Difference sampl/calibration port	0,50	%	32	0,02	0,00
21	Uncertainty in calibration gas	3,0	% of limit value	31	0,13	0,02

[a] calculated by extrapolation from the test results at 19 mmol/mol (see Equation F.11)

Table F.14 — Result of the calculation

combined standard uncertainty	0,625	µmol/mol
expanded uncertainty	1,25	µmol/mol
expanded uncertainty actual	14,5	%
expanded uncertainty required	15	%

CONCLUSION: $U_{c,\,rel} \leq U_{req,\,rel}$

Requirement 2 is fulfilled.

F.5 Type approval requirement 3

Table F.15 gives the performance characteristics that shall be considered in demonstrating compliance with requirement 3 in the type approval procedure.

Table F.15 — Relevant performance characteristics and criteria

No.	Performance characteristic	Symbol	Clause	Performance criterion for CO			Results field test	[b]
10	Reproducibility standard deviation under field conditions	$s_{r,f}$	8.5.5	≤	5,0	% of the average of a three month period	1,1	✓
11	Long term drift at zero	$D_{l,z}$	8.5.4	≤	0,5	µmol/mol	0,087	✓
12	Long term drift at span level [a]	$D_{l,s}$	8.5.4	≤	5,0	% of maximum of certification range	2,0	✓
19	Period of unattended operation		8.5.6	>	3	months or less if manufacturer indicates a shorter period	4	✓
20	Availability of the analyser	A_a	8.5.7	>	90	%	93	✓

[a] span level is 70 % to 80 % of the certification range

[b] ✓ : requirement is met

[c] if relevant

CONCLUSION: All values of the performance characteristics obtained in the field tests are complying with the requirements.

Requirement 3 in is fulfilled.

F.6 Type approval requirement 4

The values of the following uncertainties shall be included in the calculation of the expanded uncertainty after the laboratory and field tests:

Table F.16 — Standard uncertainties to be incorporated in the calculation of the expanded uncertainty after the laboratory and field tests

No.	Standard uncertainty due to	Symbol	Equation, F.
1	Repeatability at zero	$u_{r,z}$	6, 7
2	Repeatability at the 8-hour mean limit value [a]	$u_{r,lv}$	8, 9, 10
3	Lack of fit at the 8-hour mean limit value	$u_{l,lv}$	11
4	Variation in sample gas pressure at the 8-hour mean limit value	u_{gp}	12, 13
5	Variation in sample gas temperature at the 8-hour mean limit value	u_{gt}	14, 15

No.	Standard uncertainty due to	Symbol	Equation, F.
6	Variation in surrounding temperature at the 8-hour mean limit value	u_{st}	16, 17
7	Variation in electrical voltage at the 8-hour mean limit value	u_V	18, 19
8	Interferents		
8a	H_2O with concentration 21 mmol/mol	u_{H2O}	20, 21, 22, 23
8b	CO_2 with concentration 500 µmol/mol	u_{CO2}	24, 25
8c	NO with concentration 1 µmol/mol	u_{NO}	24, 25
8d	N_2O with concentration 50 nmol/mol	u_{N2O}	24, 25
9	Averaging error	u_{av}	30
10	Reproducibility under field conditions [a]	$u_{r,f}$	34
11	Long term drift at zero	$u_{d,l,z}$	35
12	Long term drift at span level	$u_{d,l,lv}$	36
18	Difference sample/calibration port	u_{Dsc}	32

[a] Take for the calculation of the combined standard uncertainty the repeatability at the 8-hour mean limit value or the reproducibility under field conditions, whichever is the greatest.

Next to the uncertainties due to these performance characteristics the uncertainty in the concentration of calibration gas and in the calibration itself are incorporated in the uncertainty calculation.

Table F.17 — Standard uncertainty of the calibration gas to be incorporated in the calculation of the expanded uncertainty after the laboratory and field tests

No.	Standard uncertainty due to	Symbol	Equation, F.
20	Uncertainty in calibration gas	u_{cg}	31

The calculation of standard uncertainties is based on the procedures laid down in EN ISO 1495. The uncertainty calculation shall be carried out with the values of the performance characteristics at the concentration of the 8-hour mean limit value (if relevant).

The absolute **expanded uncertainty**, U_c, shall be calculated according to:

$$U_c = k \times u_c \tag{F.1}$$

with k = 2

where

U_c is the expanded uncertainty (µmol/mol);

k is the coverage factor of approximately 95 %;

u_c is the combined standard uncertainty (µmol/mol);

The relative **expanded uncertainty**, $U_{c,rel}$, shall be calculated according to:

$$U_{c,rel} = (U_c / hlv) \times 100\,\% \qquad (F.2)$$

where

$U_{c,rel}$ is the relative expanded uncertainty (%);

U_c is the expanded uncertainty (µmol/mol);

hlv is the 8-hour mean limit value (µmol/mol);

Requirement 4 is fulfilled when: $U_{c,rel} \leq U_{req,rel}$

The **combined standard uncertainty**, u_c, is calculated using the formula:

$$u_c = \sqrt{u_{r,z}^2 + (u_{r,lv}^2 \text{ or } u_{r,f}^2) + u_l^2 + u_{gp}^2 + u_{gt}^2 + u_{st}^2 + u_V^2 + u_{H_2O}^2 + (u_{int,pos}^2 \text{ or } u_{int,neg}^2) + u_{av}^2 + u_{d,l,z}^2 + u_{d,l,lv}^2 + u_{D_{sc}}^2 + u_{cg}^2} \qquad (F.33)$$

with

$u_{r,z}$ is the standard uncertainty for repeatability at zero (µmol/mol);

$u_{r,lv}$ is the standard uncertainty for repeatability at the 8-hour mean limit value (nµmol/mol);

u_l is the standard uncertainty for lack of fit at the 8-hour mean limit value (µmol/mol);

u_{gp} is the standard uncertainty for sample gas pressure variation (µmol/mol);

u_{gt} is the standard uncertainty for sample gas temperature variation (µmol/mol);

u_{st} is the standard uncertainty for surrounding temperature variation (µmol/mol);

u_V is the standard uncertainty for electrical voltage variation (µmol/mol);

u_{H2O} is the standard uncertainty for the presence of water vapour (µmol/mol);

$u_{int,pos}$ is the standard uncertainty for interferents (except water vapour) with positive impact (µmol/mol);

$u_{int,neg}$ is the standard uncertainty for interferents (except water vapour) with negative impact (µmol/mol);

u_{av} is the standard uncertainty for averaging (µmol/mol);

$u_{r,f}$ is the standard uncertainty for reproducibility under field conditions (µmol/mol);

$u_{d,l,z}$ is the standard uncertainty for long term drift at zero (µmol/mol);

$u_{d,l,lv}$ is the standard uncertainty for long term drift at level of the 8-hour mean limit value (µmol/mol);

u_{Dsc} is the standard uncertainty for difference sample/calibration port (µmol/mol);

u_{cg} is the standard uncertainty for calibration gas (µmol/mol).

For the calculation of the standard uncertainties the test values as given in F.4 have been used.

For the calibration frequency the following value has been used:

Calibration frequency: 1 x / 3 month

The standard uncertainty for **repeatability at zero**, $u_{r,z}$, is calculated according to Equation (F.6, F.7).

The standard uncertainty for **repeatability at the 8-hour mean limit value**, $u_{r,lv}$, is calculated according to Equations (F.8, F.9, F.10)

The standard uncertainty due to **lack of fit at the 8-hour mean limit value**, u_l, is calculated according to Equation (F.11).

The standard uncertainty due to variation of **sample gas pressure at the 8-hour mean limit value**, u_{sp}, is calculated according to Equations (F.12, F.13).

The standard uncertainty due to variation of **sample gas temperature at the 8-hour mean limit value**, u_{gt}, is calculated according to Equations (F.14, F.15).

The standard uncertainty due to variation of **surrounding temperature at the 8-hour mean limit value**, u_{gt}, is calculated according to Equations (F.16, F.17).

The standard uncertainty due to variation of **electrical voltage at the 8-hour mean limit value**, u_V, is calculated according to Equations (F.18, F.19).

The standard uncertainty due to **interference by the presence of water vapour**, u_{H2O}, is calculated according to Equations (F.20, F.21, F.22, F.23).

According to EN ISO 14956 the summed uncertainties due to the interferents with positive impact and the summed uncertainties of the interferents with negative impact shall be calculated. Take the highest sum as the representative value for all interferents. See Equations (F.24, F.25, F.26, F.27, F.28, F.29).

The standard uncertainty due to the **averaging error**, u_{av}, is calculated according to Equation (F.30).

The standard uncertainty due to the **reproducibility under field conditions**, $u_{r,f}$, is calculated according to:

$$u_{r,f} = hlv \text{ x } (s_{r,f}/100) \tag{F.34}$$

where

$u_{r,f}$ is the standard uncertainty due to reproducibility under field conditions (µmol/mol);

hlv is the hourly limit value (µmol/mol);

$s_{r,f}$ is the standard uncertainty for reproducibility under field conditions (%);

The standard uncertainty due the **long term drift at zero**, $u_{d,l,z}$, is calculated according to:

$$u_{d,l,z} = D_{l,z} / \sqrt{3} \tag{F.35}$$

where

$u_{d,l,z}$ is the uncertainty due to long term drift at zero (µmol/mol);

$D_{l,z}$ is the long term drift at zero (µmol/mol).

The standard uncertainty due to the **long term drift at level of the 8-hour mean limit value**, $u_{d,l,lv}$, is calculated according to:

$$u_{d,l,lv} = (D_{l,lv} / 100) \times hlv) / \sqrt{3} \tag{F.36}$$

where

$u_{d,l,lv}$ is the uncertainty due to long term drift at the 8-hour mean limit value (µmol/mol);

$D_{l,lv}$ is the long term drift at level of the 8-hour mean limit value (%);

hlv is the 8-hour mean limit value (µmol/mol).

The standard uncertainty due to the **calibration gas**, u_{cg}, is calculated according to Equation (F.31).

The standard uncertainty due to the **difference sample/calibration port**, u_{Dsc}, is calculated according to Equation (F.32).

F.7 Example Result of type approval for CO with the results of the laboratory and field tests

F.7.1 Values of test parameters to be used in the uncertainty calculations

Next to the values given in F.4.1 the following values are used.

Table F.18 — Test concentration of the measurand

Average concentration over period of long term drift	0,10	µmol/mol

Table F.19 — Default value calibration frequency

Calibration frequency	1x / 3 months

F.7.2 Uncertainty due to interferents, except water vapour (see EN ISO 14956:2002, 8.5.5)

See F.4.2.

F.7.3 Calculation of expanded uncertainty from the results of laboratory and field tests for CO

Table F.20 — Uncertainty from the results of laboratory and field tests

No.	Standard uncertainty due to	Result or default value		Equation F.	uncert. u(p) µmol/mol	uncert. $u(p)^2$ $(\mu mol/mol)^2$
1	Repeatability standard deviation at zero	0,40	µmol/mol	6, 7	0,02	0,00
2	Repeatability standard deviation at the 8-hour mean limit valuespan level	2,0	µmol/mol	8, 9, 10	0,19 [b]	0,04
3	Lack of fit at the 8-hour mean limit value	2,5	% of the tested value	11	0,12	0,02
4	Sensitivity coefficient of sample gas pressure	0,12	µmol/mol/kPa	12, 13	0,27	0,08
5	Sensitivity coefficient of sample gas temperature	0,10	µmol/mol/K	14, 15	0,23	0,05
6	Sensitivity coefficient of surrounding temperature	0,10	µmol/mol/K	16, 17	0,23	0,05
7	Sensitivity coefficient of electrical voltage	0,05	µmol/mol/V	18, 19	0,15	0,02
8	Interferents					
	8a H_2O with concentration 21 mmol/mol [a]	0,33	µmol/mol	20, 21, 22, 23	0,23	0,05
	8b – 8d summed uncertainty with positive or negative impact (whichever is the greatest)			28, 29	0,20	0,04
9	Averaging effect	3,0	% of the tested value	30	0,15	0,02
18	Difference sampl/calibration port	0,50	%	32	0,02	0,00
10	Reproducibility standard deviation under field conditions	1,1	% of average of 3-months period	34	0,09 [a]	0,01
11	Long term drift at zero	0,087	µmol/mol	35	0,05	0,00
12	Long term drift at limit value	2,0	% of limit value	36	0,10	0,01
21	Uncertainty in calibration gas	3,0	% of limit value	31	0,13	0,02

[a] calculated by extrapolation from the test results at 19 mmol/mol (see Equation F.11)

[b] Take for the calculation of the combined standard uncertainty the repeatability at the 8-hour mean limit value or the reproducibility under field conditions, whichever is the greatest.

Table F.21 — Result of the calculation

combined standard uncertainty	0,635	µmol/mol
expanded uncertainty	1,27	µmol/mol
expanded uncertainty actual	14,7	%
expanded uncertainty required	15	%

CONCLUSION: $U_{c,\ rel} \leq U_{req,\ rel}$

Requirement 4 is fulfilled.

Annex G
(normative)

Calculation of uncertainty in field operation at the 8-hour mean limit value

G.1 Uncertainty in hourly measurements at the level of the 8-hour mean limit value

G.1.1 Combined standard uncertainty

The **combined standard uncertainty**, u_c, in an hourly measurement at the 8-hour mean limit value during field operation of a measurement system shall be calculated using the following equation:

$$u_{\mathrm{c}} = \sqrt{u_{\mathrm{r,z}}^2 + (u_{\mathrm{r,lv}}^2 \text{ or } u_{\mathrm{r,f}}^2) + u_{\mathrm{l}}^2 + u_{\mathrm{gp}}^2 + u_{\mathrm{gt}}^2 + u_{st}^2 + u_{\mathrm{v}}^2 + u_{\mathrm{H2O,act}}^2 + (u_{\mathrm{int,act,pos}}^2 \text{ or } u_{\mathrm{int,act,neg}}^2) + u_{\mathrm{av}}^2 + u_{\mathrm{Dsc}}^2 + u_{\mathrm{d,l,z}}^2 + u_{\mathrm{d,l,lv}}^2 + u_{\mathrm{cg}}^2} \quad \text{(G.1)}$$

where

$u_{r,z}$ standard uncertainty for repeatability at zero (µmol/mol);

$u_{r,lv}$ standard uncertainty for repeatability at the 8-hour mean limit value (µmol/mol). The highest value of $u_{r,lv}$ and $u_{r,lv}$ shall be taken in the uncertainty calculation;

u_l standard uncertainty for lack of fit at the 8-hour mean limit value (µmol/mol);

u_{gp} standard uncertainty for sample gas pressure variation (µmol/mol);

u_{gt} standard uncertainty for sample gas temperature variation (µmol/mol);

u_{st} standard uncertainty for surrounding temperature variation (µmol/mol);

u_V standard uncertainty for electrical voltage variation (µmol/mol);

$u_{H2O,act}$ standard uncertainty for the presence of water vapour (µmol/mol);

$u_{int,act,pos}$ standard uncertainty for interferents (except water vapour) with positive impact (µmol/mol). The highest value of $u_{int,act,pos}$ and $u_{int,act,neg}$ shall be taken in the uncertainty calculation;

$u_{int,act,neg}$ standard uncertainty for interferents (except water vapour) with negative impact (µmol/mol). The highest value of $u_{int,act,pos}$ and $u_{int,act,neg}$ shall be taken in the uncertainty calculation;

u_{av} standard uncertainty for averaging (µmol/mol);

u_{Dsc} standard uncertainty for difference sample/calibration port (µmol/mol);

$u_{r,f}$ standard uncertainty for reproducibility under field conditions (µmol/mol). The highest value of $u_{r,lv}$ and $u_{r,lv}$ shall be taken in the uncertainty calculation;

$u_{d,l,z}$ standard uncertainty for long-term drift at zero (µmol/mol);

$u_{d,l,lv}$ standard uncertainty for long-term drift at level of the 8-hour mean limit value (µmol/mol);

u_{cg} standard uncertainty for calibration gas (µmol/mol).

The absolute **expanded uncertainty**, U_c, shall be calculated according to:

$$U_c = k \times u_c \quad \text{(G.2)}$$

with k = 2

where

U_c is the expanded uncertainty (µmol/mol);

k is the coverage factor of approximately 95 %;

u_c is the combined standard uncertainty (µmol/mol).

The relative **expanded uncertainty**, $U_{c,rel}$, shall be calculated according to:

$$U_{c,rel} = (U_c / hlv) \times 100 \qquad \text{(G.3)}$$

where

$U_{c,rel}$ is the relative expanded uncertainty (%);

U_c is the expanded uncertainty (µmol/mol);

hlv is the 8-hour mean limit value (µmol/mol).

G.1.2 Standard uncertainties

The standard uncertainties shall be calculated with the equations given in Annex F, using the relevant values of the performance characteristics, the values of the site-specific conditions related to physical and chemical influences, the value of the site-specific conditions related to operational parameters and the actual values of test gas concentrations during the type approval test.

The standard uncertainty for **repeatability at zero**, $u_{r,z}$, is calculated according to Equations (F.6, F.7).

The standard uncertainty for **repeatability at the 8-hour mean limit value**, $u_{r,lv}$, is calculated according to Equations (F.8, F.9, F.10)

The standard uncertainty due to **lack of fit at the 8-hour mean limit value**, u_l, is calculated according to Equation (F.11).

The standard uncertainty due to variation of **sample gas pressure at the 8-hour mean limit value**, u_{sp}, is calculated according to:

$$u_{gp} = ((hlv/c_t) \times b_{gp})(\Delta_{gp} / \sqrt{3}) \qquad \text{(G.4)}$$

$$\Delta_{gp} = P_1 - P_2 \qquad \text{(G.5)}$$

where

u_{gp} is the standard uncertainty due to the influence of pressure (µmol/mol);

hlv is the 8-hour mean limit value (µmol/mol);

c_t is the test gas concentration (70 % to 80 % of the certification range) (µmol /mol);

b_{gp} is the sensitivity coefficient of sample gas pressure variation (µmol/mol/kPa);

Δ_{gp} is the pressure range of the variation of the sample gas pressure (kPa);

P_1 is the upper level of the site-specific range of the variation of the sample gas pressure (kPa) (see G.2.3);

P_2 is the lower level of the site-specific range of the variation of the sample gas pressure (kPa) (see G.2.3).

The standard uncertainty due to variation of **sample gas temperature at the** 8-hour mean limit value, u_{gt}, is calculated according to:

$$u_{gt} = ((hlv/c_t) \times b_{gt})(\Delta_{gt}/\sqrt{3}) \tag{G.6}$$

$$\Delta_{gt} = T_{gt,1} - T_{gt,2} \tag{G.7}$$

where

u_{gt} is the standard uncertainty due to the variation of sample gas temperature (µmol/mol);

hlv is the 8-hour mean limit value (µmol/mol);

c_t is the test gas concentration (70 % to 80 % of the certification range) (µmol/mol);

b_{gt} is the sensitivity coefficient of sample gas temperature variation (µmol/mol);

Δ_{gt} is the sample gas temperature range of the variation of the sample gas temperature (K);

$T_{gt,1}$ is the upper level of the site-specific range of the variation of the sample gas temperature (°C) (see G.2.3);

$T_{gt,2}$ is the lower level of the site-specific range of the variation of the sample gas temperature (°C) (see G.2.3).

The standard uncertainty due to variation of **surrounding temperature at the** 8-hour mean limit value, u_{st}, is calculated according to:

$$u_{st} = ((hlv/c_t) \times b_{st})(\Delta_{st}/\sqrt{3}) \tag{G.8}$$

$$\Delta_{st} = T_{st,1} - T_{st,2} \tag{G.9}$$

where

u_{st} is the standard uncertainty due to the variation of the surrounding temperature (µmol/mol);

hlv is the 8-hour mean limit value (µmol/mol);

c_t is the test gas concentration (70 % to 80 % of the certification range) (µmol/mol);

b_{st} is the sensitivity coefficient of the surrounding temperature variation (µmol/mol/K);

Δ_{st} is the surrounding temperature range of the variation of the surrounding temperature (K);

$T_{st,1}$ is the upper level of the site-specific range of the variation of the surrounding temperature (°C) (see G.2.3);

$T_{st,2}$ is the lower level of the site-specific range of the variation of the surrounding temperature (°C) (see G.2.3).

The standard uncertainty due to variation of **electrical voltage at the** 8-hour mean limit value, u_V, is calculated according to:

$$u_v = ((hlv/c_t) \times b_v)(\Delta_v/\sqrt{3}) \quad \text{(G.10)}$$

$$\Delta_v = V_1 - V_2 \quad \text{(G.11)}$$

where

u_V is the standard uncertainty due to the variation of electrical voltage (µmol/mol);

hlv is the 8-hour mean limit value (µmol/mol);

c_t is the test gas concentration (70 % to 80 % of the certification range) (µmol/mol);

b_V is the sensitivity coefficient of electrical voltage variation (µmol/mol/V);

Δ_V is the electrical voltage range of the variation of the electrical voltage (V);

V_1 is the upper level of the site-specific range of the variation of the electrical voltage (V) (see G.2.3);

V_2 is the lower level of the site-specific range of the variation of the electrical voltage (V) (see G.2.3).

The calculation of the uncertainty due to interferents has to based on the actual concentration of the chemical interferents during field operation. Therefore the following equations have to be used for calculating the uncertainty due to water vapour and other other chemical interfering compounds.

The standard uncertainty due to the **presence of water vapour at the 8-hour mean limit value**, $u_{H_2O,act}$, is calculated according to:

$$X_{\mathrm{H_2O,z,act}} = (c_{\mathrm{H_2O,act,max}} / c_{\mathrm{H_2O,max}}) \times X_{\mathrm{H_2O,z,max}} \quad \text{(G.12)}$$

$$X_{\mathrm{H_2O},c_t,\mathrm{act}} = (c_{\mathrm{H_2O,act,max}} / c_{\mathrm{H_2O,max}}) \times X_{\mathrm{H_2O},c_t,\mathrm{max}} \quad \text{(G.13)}$$

$$X_{\mathrm{H_2O,act}} = ((X_{\mathrm{H_2O},c_t,\mathrm{act}} - X_{\mathrm{H_2O,z,act}}) / c_t) \times hlv + X_{\mathrm{H_2O,z,act}} \quad \text{(G.14)}$$

$$u_{\mathrm{H_2O,act}} = \left|(X_{\mathrm{H_2O,act}} / c_{\mathrm{H_2O,act,max}})\right| \times \sqrt{((c_{\mathrm{H_2O,act,max}}{}^2 + c_{\mathrm{H_2O,act,max}} \times c_{\mathrm{H_2O,act,min}} + c_{\mathrm{H_2O,act,min}}{}^2)/3} \quad \text{(G.15)}$$

where

$X_{\mathrm{H_2O,z,act}}$ is the influence quantity of the actual H_2O concentration at zero concentration of the measurand (µmol/mol);

$X_{H_2O,c_t,act}$ is the influence quantity of the actual H_2O concentration at the test concentration c_t of the measurand (µmol/mol);

$c_{H_2O,\max}$ is the maximum concentration of water vapour used in the calculations for type approval (= 21 mmol/mol);

$X_{H_2O,z,\max}$ is the influence quantity of an H_2O concentration of 21 mmol/mol at zero concentration of the measurand (µmol/mol) (calculated with Equation F.20);

$X_{H_2O,c_t,\max}$ is the influence quantity of an H_2O concentration of 21 mmol/mol at the test concentration c_t of the measurand (µmol/mol) (calculated with Equation F.21);

$X_{H_2O,act}$ is the influence quantity of the actual H_2O concentration at the 8-hour mean limit value (µmol/mol);

c_t is the test gas concentration of the measurand (µmol/mol);

hlv is the 8-hour mean limit value (µmol/mol);

$u_{H_2O,act}$ is the standard uncertainty due to interference by the presence of water vapour at the actual concentration of water vapour (µmol/mol);

$c_{H_2O,act,\max}$ is the actual maximum concentration of water vapour (mmol/mol);

$c_{H_2O,act,\min}$ is the actual minimum concentration of water vapour (mmol/mol).

The standard uncertainty due to each **interfering compound at the 8-hour mean limit value** (other than water vapour), $u_{\text{int,act}}$, is calculated according to:

$$X_{\text{int,z,act}} = (c_{\text{int,act,max}} / c_{\text{int,max}}) \times X_{\text{int,z,max}} \quad \text{(G.16)}$$

$$X_{\text{int,c}_\text{t}\text{,act}} = (c_{\text{int,act,max}} / c_{\text{int,max}}) \times X_{\text{int,c}_\text{t}\text{,max}} \quad \text{(G.17)}$$

$$X_{\text{int,act}} = ((X_{\text{int,c}_\text{t}\text{,act}} - X_{\text{int,z,act}}) / c_\text{t}) \times hlv + X_{\text{int,z,act}} \quad \text{(G.18)}$$

$$u_{\text{int,act}} = \left| (X_{\text{int,act}} / c_{\text{int,act,max}}) \right| \times \sqrt{((c_{\text{int,act,max}}{}^2 + c_{\text{int,act,max}} \times c_{\text{int,act,min}} + c_{\text{int,act,min}}{}^2) / 3)} \quad \text{(G.19)}$$

where

$X_{\text{int,act,z}}$ is the influence quantity of the actual concentration (nmol/mol resp. µmol/mol) of the relevant interfering compound at zero concentration of the measurand (µmol/mol);

$X_{\text{int,act,ct}}$ is the influence quantity of the actual concentration (nmol/mol resp. µmol/mol) of the relevant interfering compound at the test concentration c_t of the measurand (µmol/mol);

$c_{\text{int,max}}$ is the maximum concentration of the relevant interfering compound used in the calculations for type approval (µmol/mol);

$X_{\text{int,max,z}}$ is the influence quantity of the maximum concentration (nmol/mol resp. µmol/mol) of the relevant interfering compound at zero concentration of the measurand (µmol/mol);

$X_{\text{int,max,ct}}$ is the influence quantity of the maximum concentration (nmol/mol resp. µmol/mol) of the relevant interfering compound at the test concentration c_t of the measurand (µmol/mol);

c_t is the test concentration of the measurand at the level of the 8-hour mean limit value (µmol/mol);

hlv is the 8-hour mean limit value (µmol/mol);

$u_{\text{int,act}}$ is the standard uncertainty at the 8-hour mean limit value due to interference by the presence of a chemical compound at the actual concentration of the compound (µmol/mol);

$c_{\text{int,act,max}}$ is the actual maximum concentration of the relevant interfering compound (µmol/mol resp. µmol/mol);

$c_{\text{int,act,min}}$ is the actual minimum concentration of the relevant interfering compound (µmol/mol resp. µmol/mol);

According to EN ISO 14956 the summed uncertainties due to the interferents with positive impact and the summed uncertainties of the interferents with negative impact shall be calculated:

$$S(u_{\text{int},act,pos}) = u_{\text{int},act,1,pos} + u_{\text{int},act,2,pos} + ... + u_{\text{int},act,n,pos} \quad \text{(G.20)}$$

$$S(u_{\text{int},act,pos}) = u_{\text{int},act,1,pos} + u_{\text{int},act,2,pos} + ... + u_{\text{int},act,n,pos} \quad \text{(G.21)}$$

Take the highest sum as the representative value for all interferents:

$$u_{\text{int},act,pos} = \sqrt{S(u_{\text{int},act,pos})^2} \tag{G.22}$$

$$u_{\text{int},act,neg} = \sqrt{S(u_{\text{int},act,neg})^2} \tag{G.23}$$

where

$u_{\text{int,act,pos}}$ is the sum of uncertainties due to interferents with positive impact (µmol/mol);

$u_{\text{int,act,1,pos}}$ is the uncertainty due to the first interferent with positive impact (µmol/mol);

$u_{\text{int,act},n,\text{pos}}$ is the uncertainty due to the n^{th} interferent with positive impact (µmol/mol);

$u_{\text{int,act,neg}}$ is the sum of uncertainties due to interferents with negative impact (µmol/mol);

$u_{\text{int,act,1,neg}}$ is the uncertainty due to the first interferent with negative impact (µmol/mol);

$u_{\text{int,act},n,\text{neg}}$ is the uncertainty due to the n^{th} interferent with negative impact (µmol/mol);

The standard uncertainty due to the **averaging error**, u_{av}, is calculated according to Equation (F.30).

The standard uncertainty due to the **reproducibility under field conditions**, $u_{\text{r,f}}$, is calculated according to Equation (F.34).

The standard uncertainty due the **long term drift at zero**, $u_{\text{d,l,z}}$, is calculated according to Equation (F.35):

The standard uncertainty due to the **long term drift at level of the 8-hour mean limit value**, $u_{\text{d,l,lv}}$, is calculated according to Equation (F.36).

The standard uncertainty due to the **calibration gas**, u_{cg}, is calculated according to Equation (F.31), with the actual uncertainty of the calibration gas.

The standard uncertainty due to the **difference sample/calibration port**, u_{Dsc}, is calculated according to Equation (F.17).

G.2 Example

G.2.1 General

To calculate the uncertainty at the 8-hour mean limit value during field operation of a measurement system the following data shall be used:

- values of the following performance characteristics obtained during the type approval test:
  - Repeatability standard deviation at zero;
  - Repeatability standard deviation at test gas concentration c_t;
  - Lack of fit;
  - Sensitivity coefficient of sample gas pressure;
  - Sensitivity coefficient of sample gas temperature;

- Sensitivity coefficient of surrounding temperature;
- Sensitivity coefficient of electrical voltage;
- Interferences at zero and test gas concentration c_t;
  - H_2O with concentration 19 mmol/mol;
  - CO_2 with concentration 500 µmol/mol;
  - NO with concentration 1 µmol/mol;
  - N_2O with concentration 50 nmol/mol;
- Averaging effect
- Difference sample/calibration port (if relevant) ;
- Reproducibility standard deviation under field conditions;
- Long term drift at zero;
- Long term drift at span level;

- values of site specific conditions related to physical influences:
  - Sample pressure variation;
  - Sample gas temperature variation;
  - Surrounding air temperature variation;
  - Voltage variation;
- values of site specific conditions related to chemical influences/interferents:
  - H_2O concentration range;
  - CO_2 concentration range;
  - NO concentration range;
  - N_2O concentration range;
- values of site specific conditions related to operational parameters:
  - Uncertainty in calibration gas;
  - Calibration frequency;
- actual values of test gas concentrations during the type approval test.

G.2.2 Values of the relevant performance characteristics obtained during the type approval test

Table G.1 gives the results of the type approval, both from the laboratory and the field tests, to be incorporated in the uncertainty calculation, together with the relevant performance criteria.

Table G.1 — Relevant performance characteristics and criteria

No.	Performance characteristic	Symbol	Clause	Performance criterion for CO	Results lab. and field test
					[a]
1	Repeatability standard deviation at zero	$s_{r,z}$	8.4.5	≤ 1,0 µmol/mol	0,40
2	Repeatability standard deviation at concentration c_t (at a level of the 8-hour mean limit value)	$s_{r,ct}$	8.4.5	≤ 3,0 µmol/mol	2,0
3	Lack of fit (at the 8-hour mean limit value)	X_l	8.4.6	≤ 5,0 % of the measured value	2,5
4	Sensitivity coefficient of sample gas pressure	b_{gp}	8.4.7	≤ 1,0 µmol/mol/kPa	0,12
5	Sensitivity coefficient of sample gas temperature	b_{gt}	8.4.8	≤ 0,30 µmol/mol/K	0,10
6	Sensitivity coefficient of surrounding temperature	b_{st}	8.4.9	≤ 0,30 µmol/mol/K	0,10
7	Sensitivity coefficient of electrical voltage	b_V	8.4.10	≤ 0,30 µmol/mol/V	0,05
8	Interferents at zero and concentration c_t (at a level of the 8-hour mean limit value) [b]		8.4.11		
8a	H_2O with concentration 19 mmol/mol [c]	$X_{H2O,z}$		≤ 1,0 µmol/mol	0,22
		$X_{H2O,ct}$		≤ 1,0 µmol/mol	0,30
8b	CO_2 with concentration 500 µmol/mol	$X_{CO2,z}$		≤ 0,50 µmol/mol	0,10
		$X_{CO2,ct}$		≤ 0,50 µmol/mol	0,15
8c	NO with concentration 1 µmol/mol	$X_{NO,z}$		≤ 0,50 µmol/mol	0,080
		$X_{NO,ct}$		≤ 0,50 µmol/mol	0,10
8d	N_2O with concentration 50 nmol/mol	$X_{N2O,z}$		≤ 0,50 µmol/mol	0,060
		$X_{N2O,ct}$		≤ 0,50 µmol/mol	0,11

No.	Performance characteristic	Symbol	Clause	Performance criterion for CO		Results lab. and field test
9	Averaging error	X_{av}	8.4.12	≤ 7,0	% of the measured value	3,0
18	Difference sample/calibration port [d]	D_{sc}	8.4.13	≤ 1,0	%	0,50
10	Reproducibility standard deviation under field conditions	$s_{r,f}$	8.5.5	≤ 5,0	% of the average of a three month period	1,1
11	Long term drift at zero	$D_{l,z}$	8.5.4	≤ 0,5	µmol/mol	0,087
12	Long term drift at span level	$D_{l,s}$	8.5.4	≤ 5,0	% of maximum of certification range	2,0
15	Response time (rise)	t_r	8.4.3	≤ 180	s	36
16	Response time (fall)	t_f	8.4.3	≤ 180	s	33

[a] to demonstrate compliance the absolute value of the performance characteristic shall be taken
[b] performance criterion is set both at zero and span level
[c] an H_2O-concentratration of 19 mmol/mol is about 80 % RH at 293 K and 101,3 kPa
[d] if relevant

NOTE µmol/mol = ppm nmol/mol = ppb

G.2.3 Values of site specific conditions related to physical influences

Table G.2 — Site specific conditions related to physical influences

	site specific			type approval		
	lower value	upper value	range	lower value	upper value	test range
Sample pressure variation	95 kPa	110 kPa	15 kPa	80 kPa	110 kPa	30 kPa
Sample gas temperature variation	0 °C	30 °C	30 K	0 °C	30 °C	30 K
Surrounding air temperature variation	10 °C	30 °C	20 K	0 °C	30 °C	30 K
Voltage variation	220 V	250 V	30 V	210 V [a]	250 V [a]	40 V [a]

[a] specified by the manufacturer

G.2.4 Values of site specific conditions related to chemical influences/interferents

Table G.3 — Site specific conditions related to chemical influences/interferents

Interferents		site specific (hourly average values)			type approval	
		to be used in uncertainty calculation during operation			used in uncertainty calculation for type approval	
		lower value, $c_{int,act,min}$	upper value, $c_{int,act,max}$		lower value, $c_{int,min}$	upper value, $c_{int,max}$
H_2O	mmol/mol	3	21		6	21
CO_2	µmol/mol	300	500		0	500
NO	µmol/mol	0	1		0	1
N_2O	nmol/mol	0	50		0	50

G.2.5 Values of site specific conditions related to operational parameters

Table G.4 — Site specific conditions related to operational parameters

	site specific	type approval
Uncertainty in calibration gas, expressed as 2 times the standard deviation	± 5 %	± 3 %
Calibration frequency	1x / 3 months	1x / 3 months

G.2.6 Actual values of test gas concentrations during the type approval test

Table G.5 — Test concentrations of the measurand

c_t at a level of the 8-hour mean limit value	9,0	µmol/mol
c_t at span level (70 % to 80 % of the certification range)	65	µmol/mol
average concentration over period of long term drift	0,10	µmol/mol

G.2.7 Result of uncertainty calculation for CO with the results of the type approval tests and the site specific conditions

Table G.6 — Result of uncertainty calculation for CO

No.	Performance characteristic	Equation	Partial uncertainty	Value of partial uncertainty µmol/mol
1	Repeatability standard deviation at zero	F.6, F.7	$u_{r,z}$	(0,40 / √(104))= 0,04
2	Repeatability standard deviation at 8-hour mean limit value	F.8, F.9, F.10	$u_{r,lv}$	(((8,6/ 9,0) • 2,0) / √(104) = 0,19
3	Lack of fit	F.11	u_l	((2,5 / 100) • 8,6) / √3 = 0,12
4	Sensitivity coefficient of sample gas pressure	G.4, G.5	u_{gp}	((8,6/ 65) • 0,12 • 15) / √3 = 0,14

No.	Performance characteristic	Equation	Partial uncertainty	Value of partial uncertainty µmol/mol
5	Sensitivity coefficient of sample gas temperature	G.6, G.7	u_{gt}	((8,6/ 65) • 0,10 • 30) / √3 = 0,23
6	Sensitivity coefficient of surrounding	G.8, G.9	u_{st}	((8,6 / 65) • 0,10 • 20) / √3 = 0,15
7	Sensitivity coefficient of electrical voltage	G.10, G.11	u_V	((8,6 / 65) • 0,15 • 30) / √3 = 0,11
8	Interferences			
8a	H_2O with actual concentration 21 mmol/mol	G.12, G.13, G.14, G.15	$u_{H2O,act}$	$\left\lvert\frac{\left(\left(\frac{(18/18)\bullet((18/16)\bullet(0{,}22))-(18/18)\bullet((18/16)\bullet(0{,}30)}{9{,}0}\right)\bullet 8{,}6+(18/18)\bullet((18/16)\bullet(0{,}22)\right)}{18}\right\rvert\bullet\sqrt{\frac{18^2+18\bullet 3+3^2}{3}}=0{,}21$
8b	CO_2 with concentration 500 µmol/mol	G.16, G.17, G.18, G.19	$u_{CO2,act}$	$\frac{\left(\left(\frac{((500/500)\bullet 0{,}15)-((500/500)\bullet 0{,}10)}{9{,}0}\right)\bullet 8{,}6+((500/500)\bullet 0{,}10)\right)}{500}\bullet\sqrt{\frac{500^2+500\bullet 300+300^2}{3}}=0{,}12$
8c	NO with concentration 1 µmol/mol	G.16, G.17, G.18, G.19	$u_{NO,act}$	$\frac{\left(\left(\frac{((1/1)\bullet 0{,}10)-((1/1)\bullet 0{,}08)}{9{,}0}\right)\bullet 8{,}6+((1/1)\bullet 0{,}08)\right)}{1}\bullet\sqrt{\frac{1^2+1\bullet 0+0^2}{3}}=0{,}06$
8d	N_2O with concentration 50 nmol/mol	G.16, G.17, G.18, G.19	$u_{N2O,act}$	$\frac{\left(\left(\frac{((50/50)\bullet(0{,}11))-((50/50)\bullet(0{,}06))}{9{,}0}\right)\bullet 8{,}6+((50/50)\bullet(0{,}06))\right)}{50}\bullet\sqrt{\frac{50^2+50\bullet 0+0^2}{3}}=0{,}06$
9	Averaging effect	F.30	u_{av}	((3,0 / 100) • 8,6) / √3 = 0,15
18	Difference sample/calibration port	F.32	u_{Dsc}	(0,50 / 100) • 8,6/√3 = 0,02
10	Reproducibility standard deviation under field	F.34	$u_{r,f}$	(1,1 /100) • 8,6 = 0,09

No.	Performance characteristic	Equation	Partial uncertainty	Value of partial uncertainty µmol/mol
	conditions			
11	Long term drift at zero	F.35	$u_{d,l,z}$	0,087 / √3 = 0,05
12	Long term drift at span level	F.36	$u_{d,l,lv}$	((2,0 / 100) • 8,6) / √3 = 0,10
20	Ucertainty in calibration gas	F.31	u_{cg}	((5,0 / 100) • 8,6) / 2 = 0,22

Compare the values of $u_{r,lv}$ and $u_{r,f}$. Take the highest one in the calculation of the combined uncertainty:

$u_{r,lv}$ = 0,19

$u_{r,f}$ = 0,09

According to EN ISO 14956 the summed uncertainties due to the interferents with positive impact and the summed uncertainties of the interferents with negative impact shall be calculated. Take the highest sum as the representative value for all interferents (see Equations (G.10) and (G.11)).

$u_{int,pos}$ = 0,12 + 0,06 + 0,06 = 0,24

$u_{int,neg}$ = 0,00

Caculate the combined uncertainty (at the 8-hour mean limit value) with Equation (G.1):

$$u_c = \sqrt{(0{,}04^2 + 0{,}19^2 + 0{,}12^2 + 0{,}14^2 + 0{,}23^2 + 0{,}15^2 + 0{,}11^2 + 0{,}10^2 + (0{,}12 + 0{,}06 + 0{,}06)^2 + 0{,}15^2 + 0{,}02^2 + 0{,}05^2 + 0{,}10^2 + 0{,}22^2)} = 0{,}56\ \mu\text{mol/mol}$$

The absolute **expanded uncertainty**, U_c, shall be calculated according to Equation (G.2):

U_c = 2 • 0,56 = 1,12 µmol/mol

The relative **expanded uncertainty**, $U_{c,rel}$, shall be calculated according to Equation (G.3):

$U_{c,rel}$ = (1,12 / 8,6) • 100 = 13,0 %

The requirement is: $\boldsymbol{U_{c,rel} \leq U_{req,rel}}$

$U_{c,rel}$ = 13,0 %

$U_{req,rel}$ = 15 %

Requirement is fulfilled.

Bibliography

[1] VIM:1993, *International vocabulary of basic and general terms in metrology*, 2nd edition, ISO.

[2] CR 14377:2002, *Air quality – Approach to uncertainty estimation for ambient air reference measurement methods*

[3] Moore, M.A. Critical Evaluation of the Analysis of Carbon Monoxide with Nondispersive Infrared, NDIR, presented at the *Ninth Conference on Methods in Air Pollution and Industrial Hygiene Studies*, Pasadena, CA, USA, Feb. 7-9, 1968.

[4] EN ISO/IEC 17025, *General requirements for the competence of testing and calibration laboratories (ISO/IEC 17025:1999)*

[5] Directive 2000/69/EC of the European Parliament and of the council of 16 November 2000 relating to limit values for benzene and carbon monoxide in ambient air

blank

BSI — British Standards Institution

BSI is the independent national body responsible for preparing British Standards. It presents the UK view on standards in Europe and at the international level. It is incorporated by Royal Charter.

Revisions

British Standards are updated by amendment or revision. Users of British Standards should make sure that they possess the latest amendments or editions.

It is the constant aim of BSI to improve the quality of our products and services. We would be grateful if anyone finding an inaccuracy or ambiguity while using this British Standard would inform the Secretary of the technical committee responsible, the identity of which can be found on the inside front cover. Tel: +44 (0)20 8996 9000. Fax: +44 (0)20 8996 7400.

BSI offers members an individual updating service called PLUS which ensures that subscribers automatically receive the latest editions of standards.

Buying standards

Orders for all BSI, international and foreign standards publications should be addressed to Customer Services. Tel: +44 (0)20 8996 9001. Fax: +44 (0)20 8996 7001. Email: orders@bsi-global.com. Standards are also available from the BSI website at http://www.bsi-global.com.

In response to orders for international standards, it is BSI policy to supply the BSI implementation of those that have been published as British Standards, unless otherwise requested.

Information on standards

BSI provides a wide range of information on national, European and international standards through its Library and its Technical Help to Exporters Service. Various BSI electronic information services are also available which give details on all its products and services. Contact the Information Centre. Tel: +44 (0)20 8996 7111. Fax: +44 (0)20 8996 7048. Email: info@bsi-global.com.

Subscribing members of BSI are kept up to date with standards developments and receive substantial discounts on the purchase price of standards. For details of these and other benefits contact Membership Administration. Tel: +44 (0)20 8996 7002. Fax: +44 (0)20 8996 7001. Email: membership@bsi-global.com.

Information regarding online access to British Standards via British Standards Online can be found at http://www.bsi-global.com/bsonline.

Further information about BSI is available on the BSI website at http://www.bsi-global.com.

Copyright

BSI
389 Chiswick High Road
London
W4 4AL